KENNESAW STATE UNIVERSITY

CS 4277

Self-supervised Learning

Dr. Chen Zhao

czhao4@kennesaw.edu

Computer Science

College of Computing and Software Engineering

Learning Objectives

- Representation Learning
- Pretext tasks from image transformations
  - Rotation, inpainting, rearrangement, coloring
  - Reconstruction-based learning (MAE)
- Contrastive representation learning
  - Intuition and formulation
  - Instance contrastive learning: SimCLR and MOCO
  - Sequence contrastive learning: CPC
  - Self-Distillation Without Labels, DINO

Representation Learning

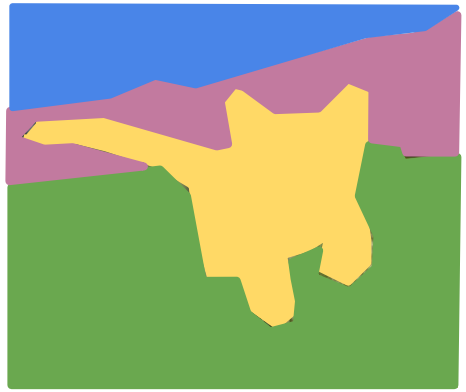
Lots of Computer Vision Tasks

Classification

CAT

No spatial extent

Semantic Segmentation



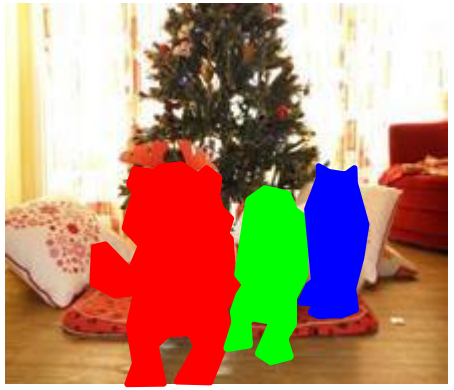
GRASS, CAT, TREE, SKY

No objects, just pixels

Object Detection

DOG, DOG, CAT

Instance Segmentation

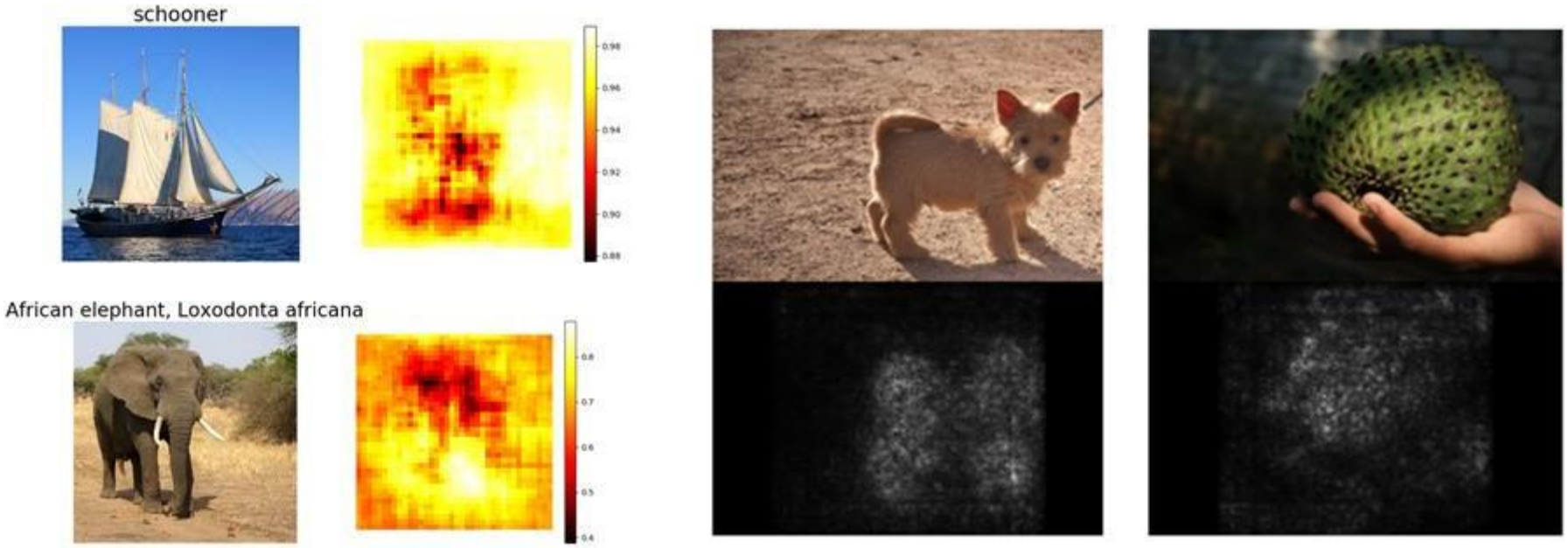


DOG, DOG, CAT

Multiple Object

This image is CC0 public domain

Visualizing and Understanding

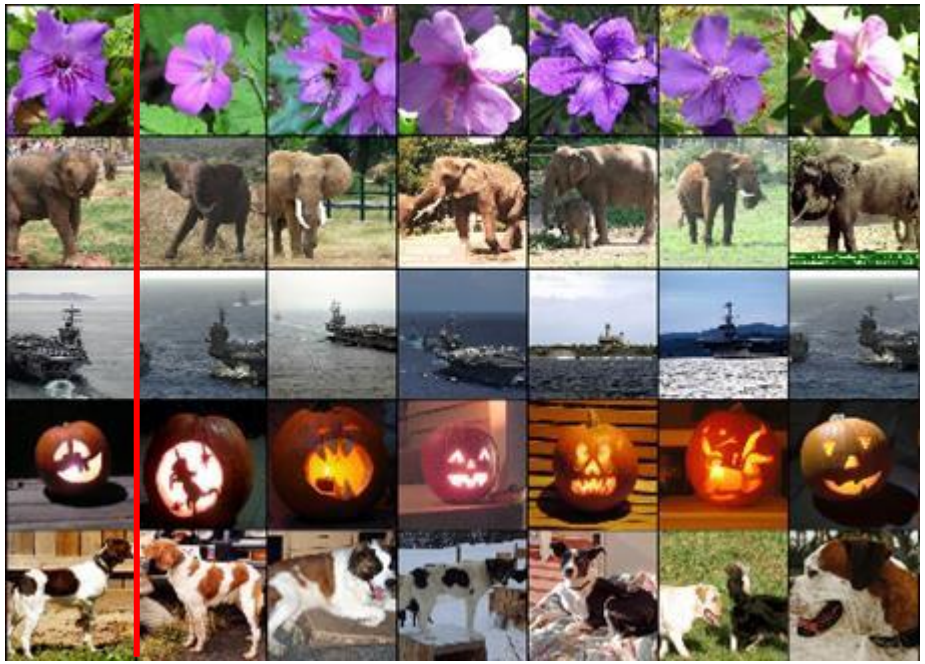


Visualizing and Understanding

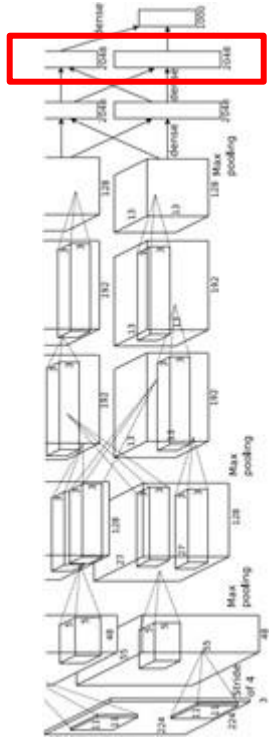
Recall: Nearest neighbors in pixel space

Test image

L2 Nearest neighbors in feature space

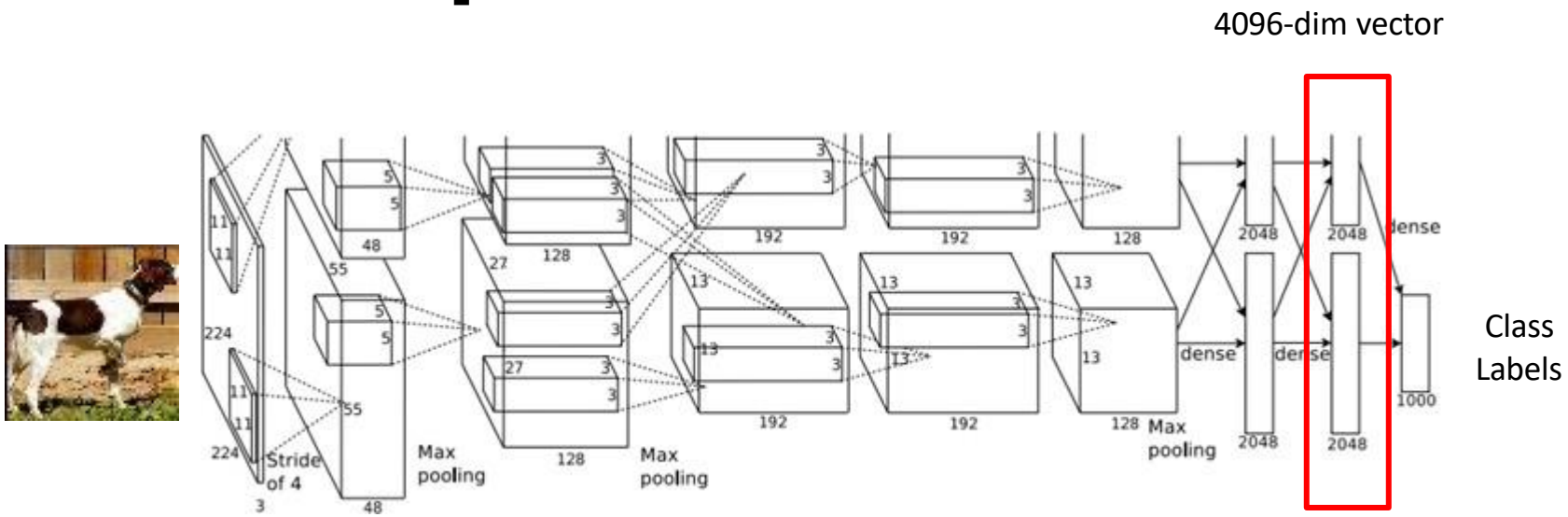


4096-dim vector



Krizhevsky et al, "ImageNet Classification with Deep Convolutional Neural Networks", NIPS 2012.
Figures reproduced with permission.

Learned Representations



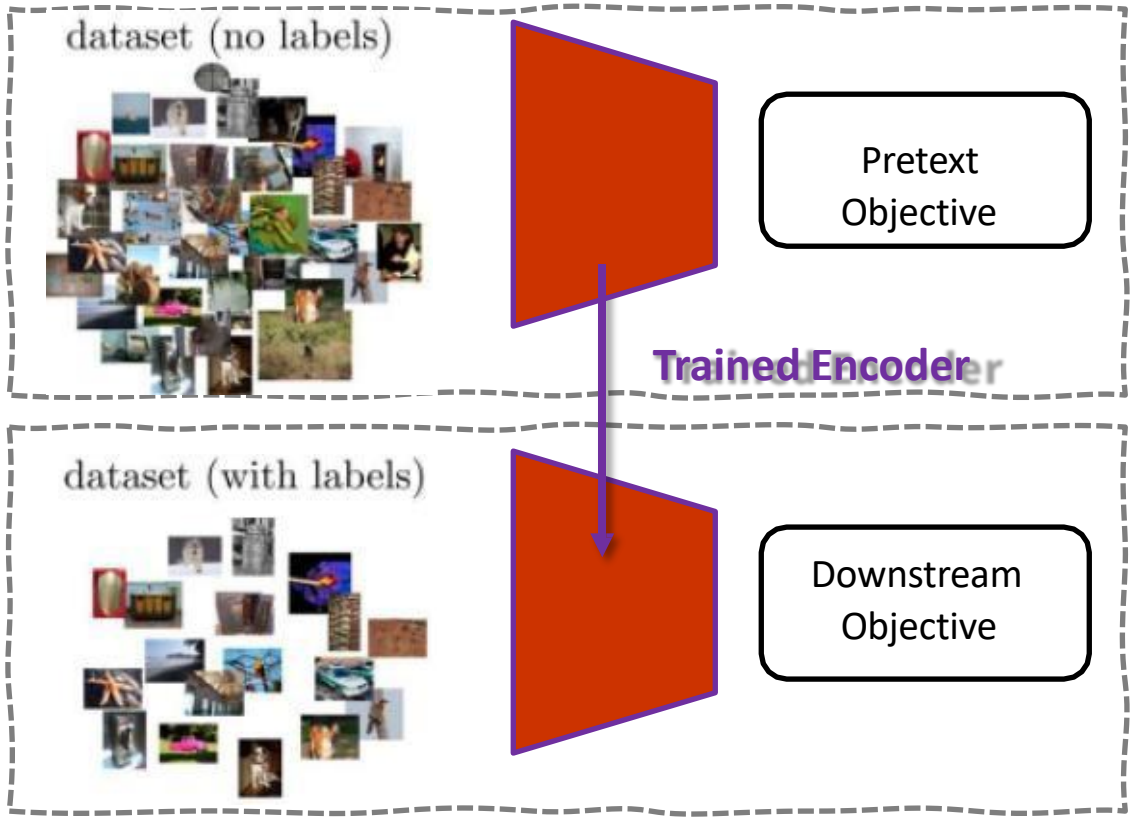
- What is the problem with large-scale training?
- - We need a lot of labeled data
- Is there a way we can train neural networks without the need for huge manually labeled datasets?

Self Supervised Learning

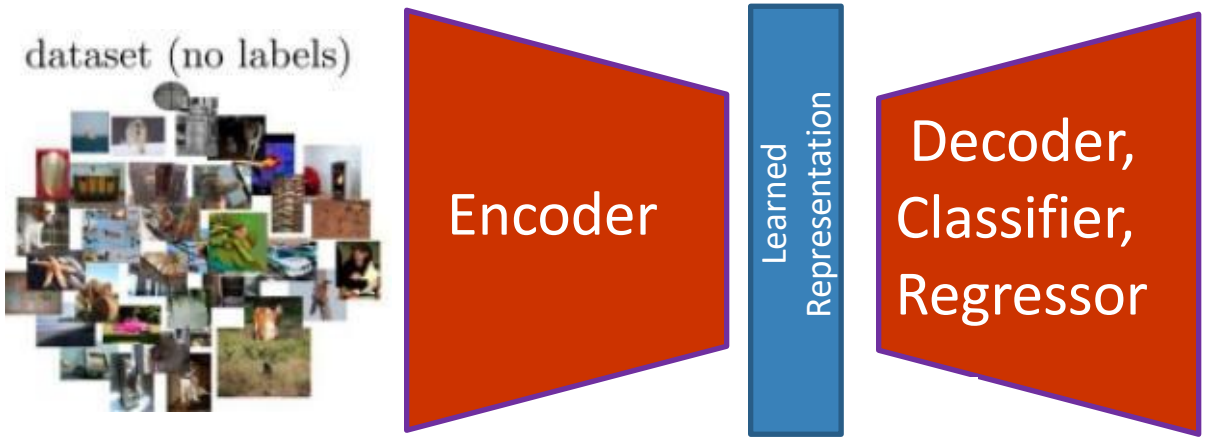
Self-Supervised Learning

- Pretext Task
  - Define a task based on the data itself
  - No manual annotation
  - Could be considered an **unsupervised** task;
  - but we learn with supervised learning objectives, e.g., classification or regression.

- Downstream Task
  - The application you care about
  - You do not have large datasets
  - The dataset is labeled

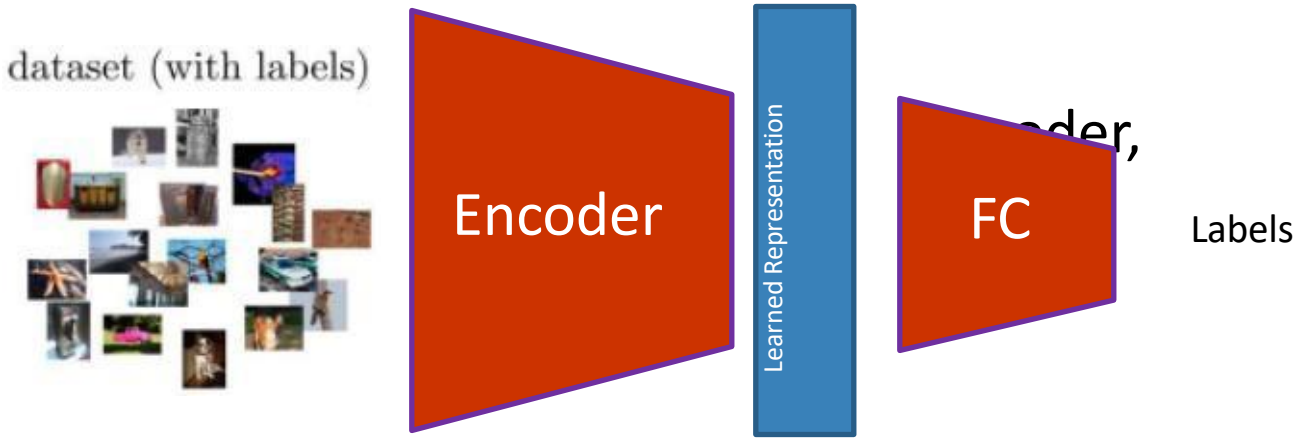


Self-Supervised Learning – Pretext Task



Labels/outputs automatically generated from data

Self-Supervised Learning – Downstream Task



Self-supervised pretext tasks

- Example: learn to predict image transformations / complete corrupted images

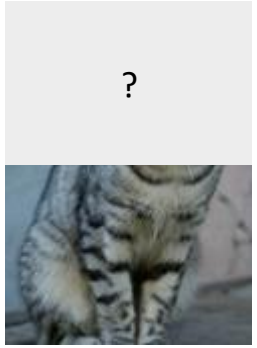
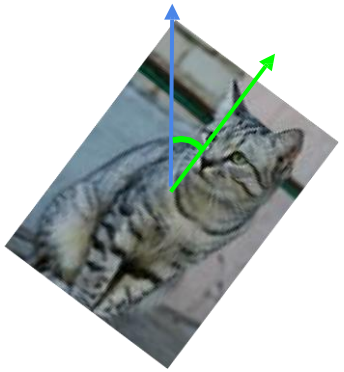


image completion



rotation prediction

“jigsaw puzzle”

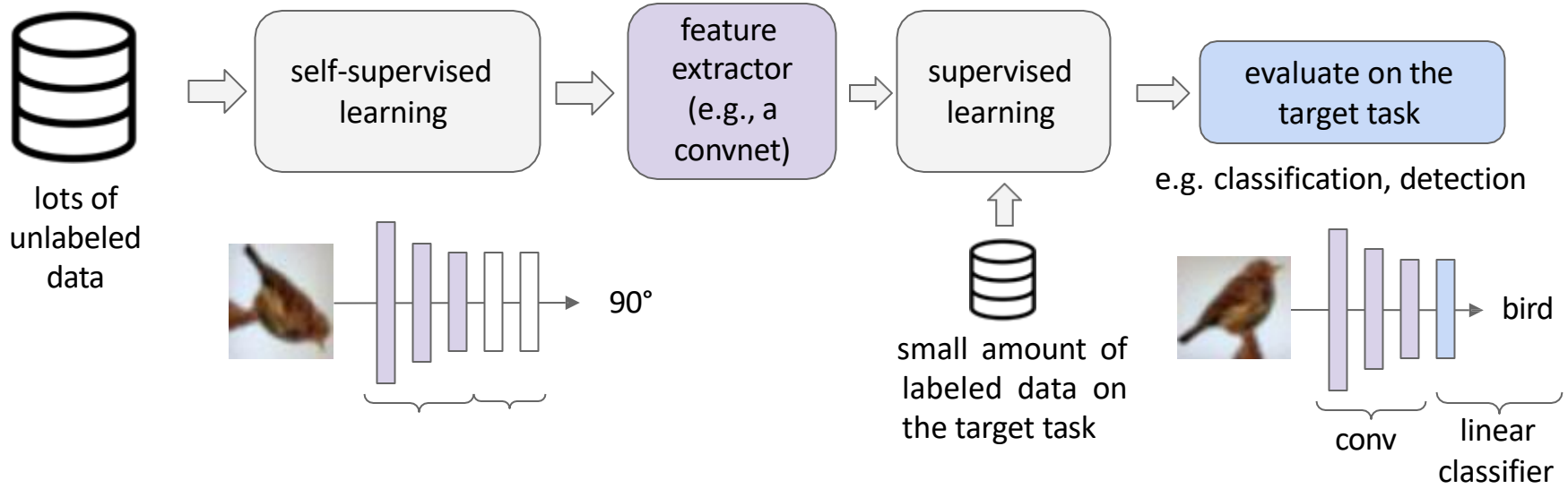
colorization

- Solving the pretext tasks allow the model to learn good features.
- We can automatically generate labels for the pretext tasks.

How to evaluate a self-supervised learning method?

- **Pretext Task Performance**
  - Measure how well the model performs on the task it was trained on without labels.
- **Representation Quality**
  - Evaluate the quality of the learned representations
  - Linear Evaluation Protocol: Train a linear classifier on the leaerned representations;
  - Clustering: Measure clustering performance;
  - t-SNE: Visualize the representations to assess their separability.)
- **Robustness and Generalization**
  - Test how well the model generalizes to different datasets and is robust to variations.
- **Computational Efficiency**
  - Assess the efficiency of the method in terms of training time and resource requirements.
- **Transfer Learning and Downstream Task Performance**
  - Assess the utility of the learned representations by transferring them to a downstream supervised task.

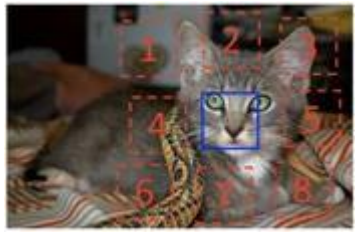
How to evaluate a self-supervised learning method?



- 1. Learn good feature extractors from self-supervised pretext tasks, e.g., predicting image rotations
- 2. Attach a shallow network on the feature extractor; train the shallow network on the target task with small amount of labeled data

Broader picture

computer vision

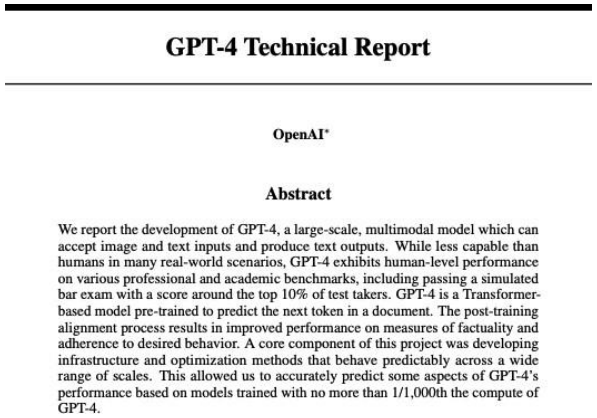


Doersch et al., 2015

robot / reinforcement learning

Dense Object Net (Florence and Manuelli et al., 2018)

language modeling



GPT-4 Technical Report

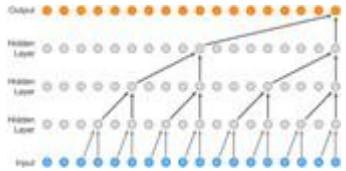
OpenAI*

Abstract

We report the development of GPT-4, a large-scale, multimodal model which can accept image and text inputs and produce text outputs. While less capable than humans in many real-world scenarios, GPT-4 exhibits human-level performance on various professional and academic benchmarks, including passing a simulated bar exam with a score around the top 10% of test takers. GPT-4 is a Transformer-based model pre-trained to predict the next token in a document. The post-training alignment process results in improved performance on measures of factuality and adherence to desired behavior. A core component of this project was developing infrastructure and optimization methods that behave predictably across a wide range of scales. This allowed us to accurately predict some aspects of GPT-4's performance based on models trained with no more than 1/1,000th the compute of GPT-4.

GPT-4 (OpenAI 2023)

speech synthesis



Wavenet (van den Oord et al., 2016)

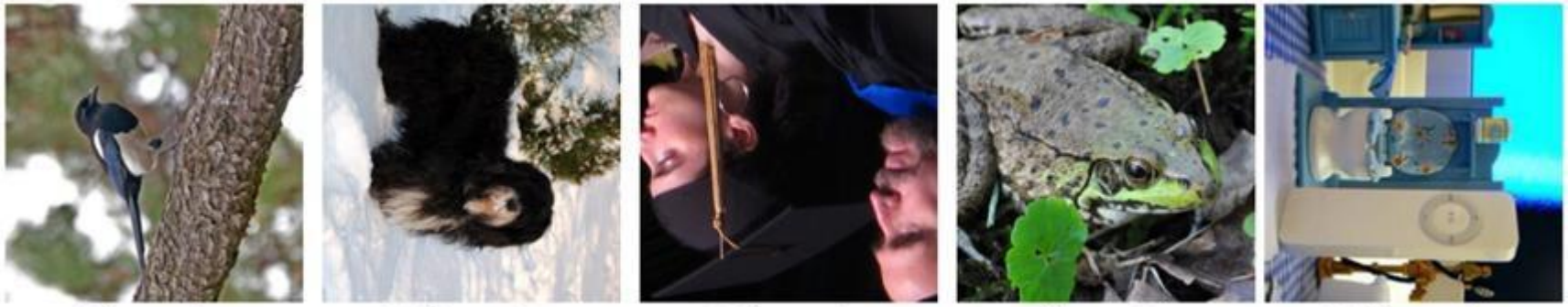
...

Pretext tasks from image transformations

Morden Pretext Tasks Approaches

- Pretext tasks from image transformations
  - Rotation, inpainting, rearrangement, coloring
  - Reconstruction-based learning (MAE)
- Contrastive representation learning
  - Intuition and formulation
  - Instance contrastive learning: SimCLR and MOCO
  - Sequence contrastive learning: CPC
  - Self-Distillation Without Labels, DINO

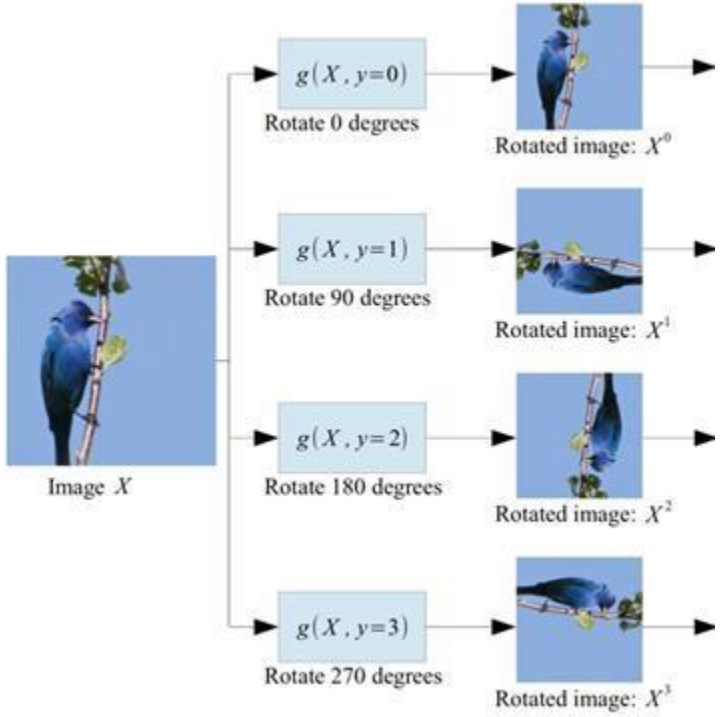
Pretext task: predict rotations



90° rotation 270° rotation 180° rotation 0° rotation 270° rotation

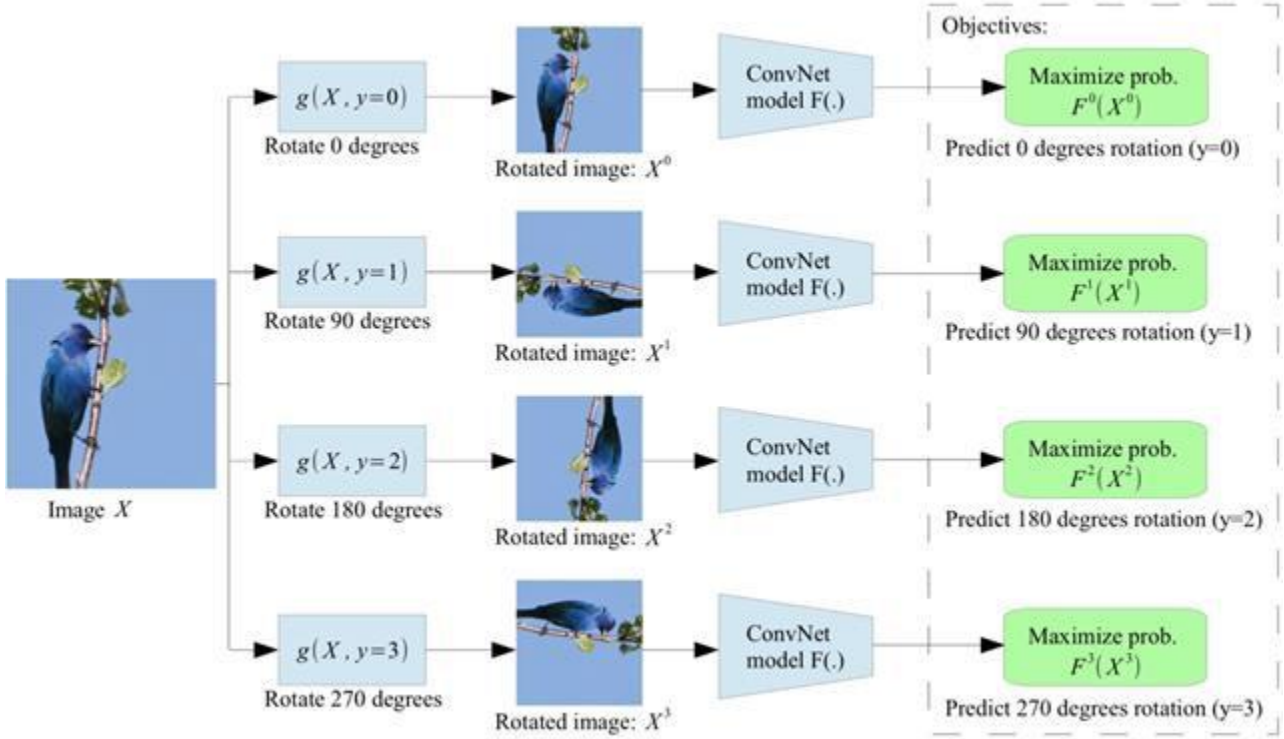
- Hypothesis: a model could recognize the correct rotation of an object only if it has the “visual commonsense” of what the object should look like unperturbed.

Pretext task: predict rotations



- Self-supervised learning by rotating the entire input images.
- The model learns to predict which rotation is applied (4-way classification)

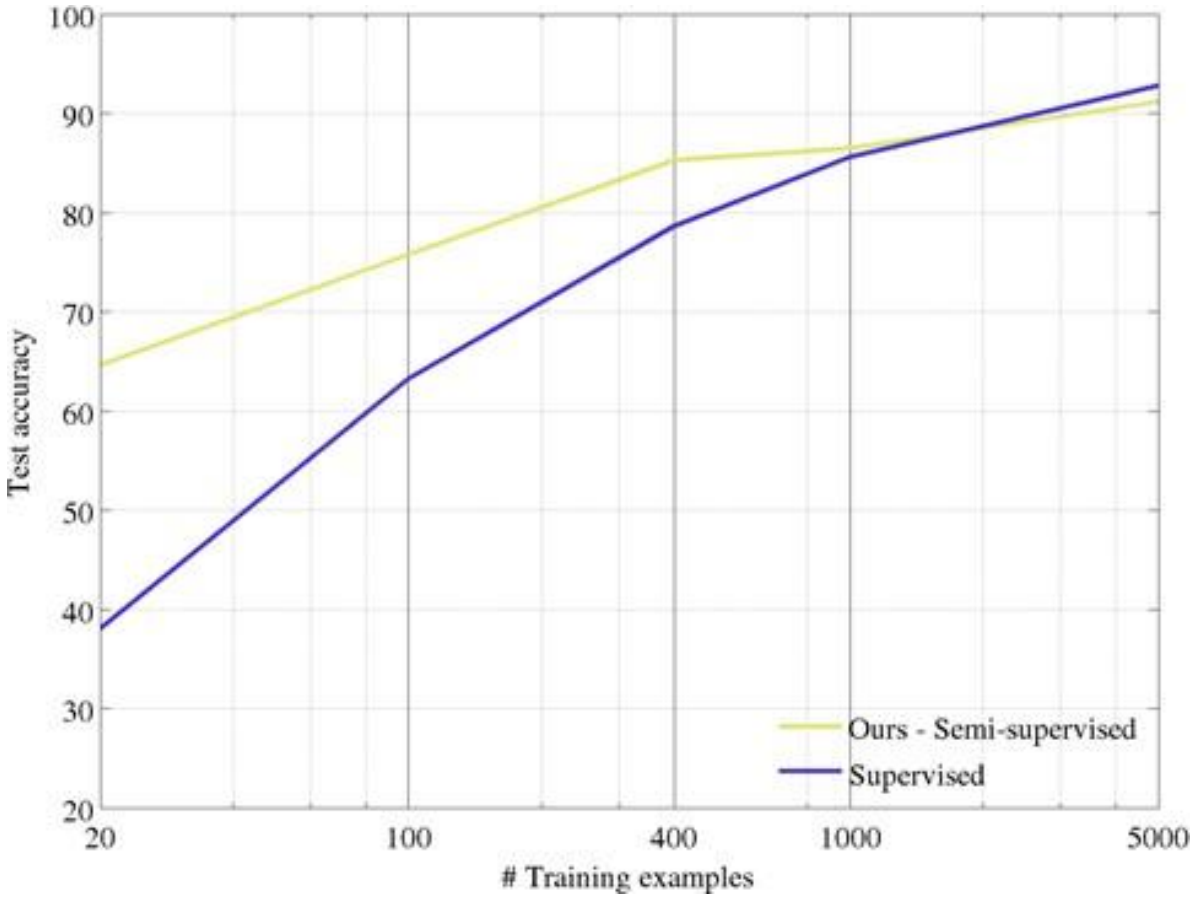
Pretext task: predict rotations



- Self-supervised learning by rotating the entire input images.
- The model learns to predict which rotation is applied (4-way classification)

Evaluation on semi-supervised learning

- Self-supervised learning on CIFAR10 (entire training set).
- Freeze conv1 + conv2
- Learn conv3 + linear layers with subset of labeled CIFAR10 data (classification).



Transfer learned features to supervised learning

	Classification (%mAP)		Detection (%mAP)	Segmentation (%mIoU)
Trained layers	fc6-8	all	all	all
ImageNet labels	78.9	79.9	56.8	48.0
Random		53.3	43.4	19.8
Random rescaled Krähenbühl et al. (2015)	39.2	56.6	45.6	32.6
Egomotion (Agrawal et al., 2015)	31.0	54.2	43.9	
Context Encoders (Pathak et al., 2016b)	34.6	56.5	44.5	29.7
Tracking (Wang & Gupta, 2015)	55.6	63.1	47.4	
Context (Doersch et al., 2015)	55.1	65.3	51.1	
Colorization (Zhang et al., 2016a)	61.5	65.6	46.9	35.6
BIGAN (Donahue et al., 2016)	52.3	60.1	46.9	34.9
Jigsaw Puzzles (Noroozi & Favaro, 2016)	-	67.6	53.2	37.6
NAT (Bojanowski & Joulin, 2017)	56.7	65.3	49.4	
Split-Brain (Zhang et al., 2016b)	63.0	67.1	46.7	36.0
ColorProxy (Larsson et al., 2017)		65.9		38.4
Counting (Noroozi et al., 2017)	-	67.7	51.4	36.6
(Ours) RotNet	**70.87**	**72.97**	**54.4**	**39.1**

Pretrained with full ImageNet supervision

No pretraining

Self-supervised learning with rotation prediction

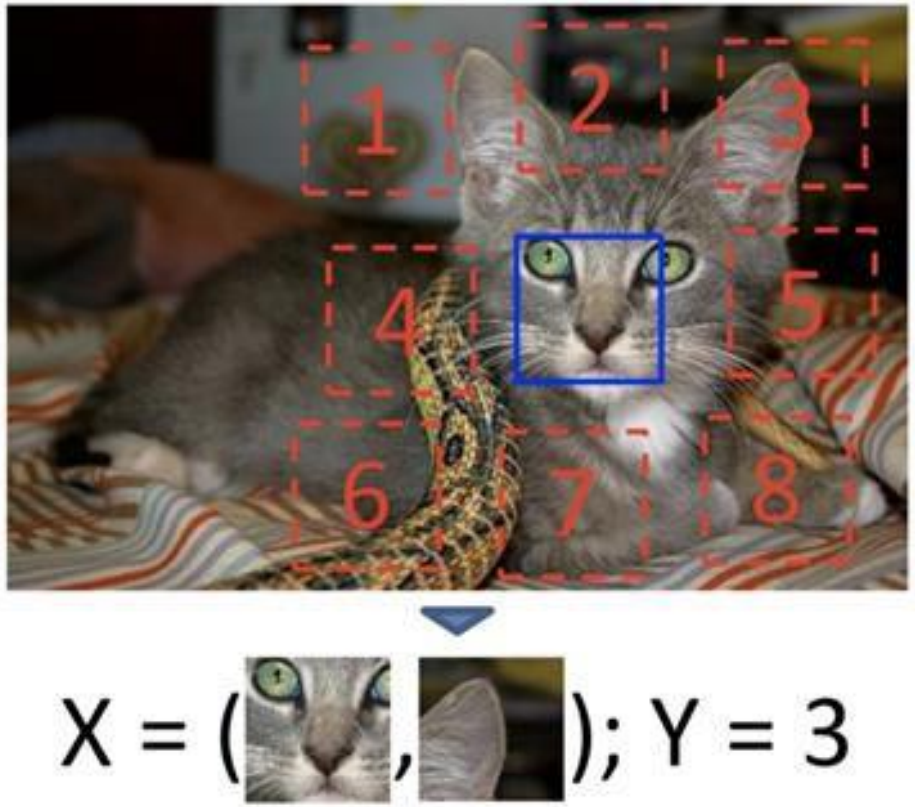
- Self-supervised learning on ImageNet (entire training set) with AlexNet.

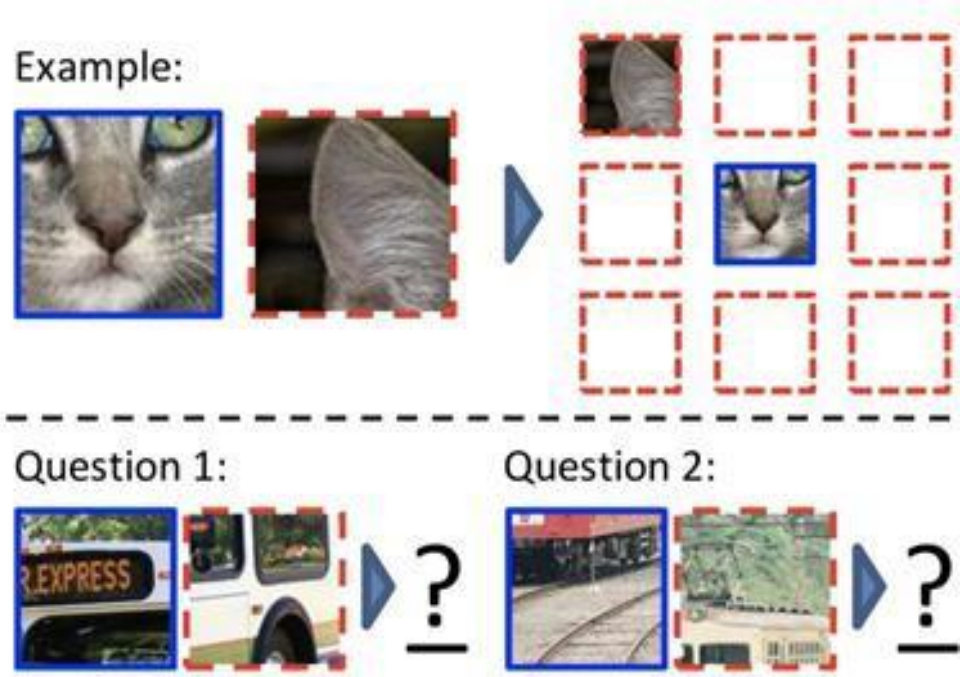
Visualize learned visual attentions

(a) **Attention maps of supervised model**

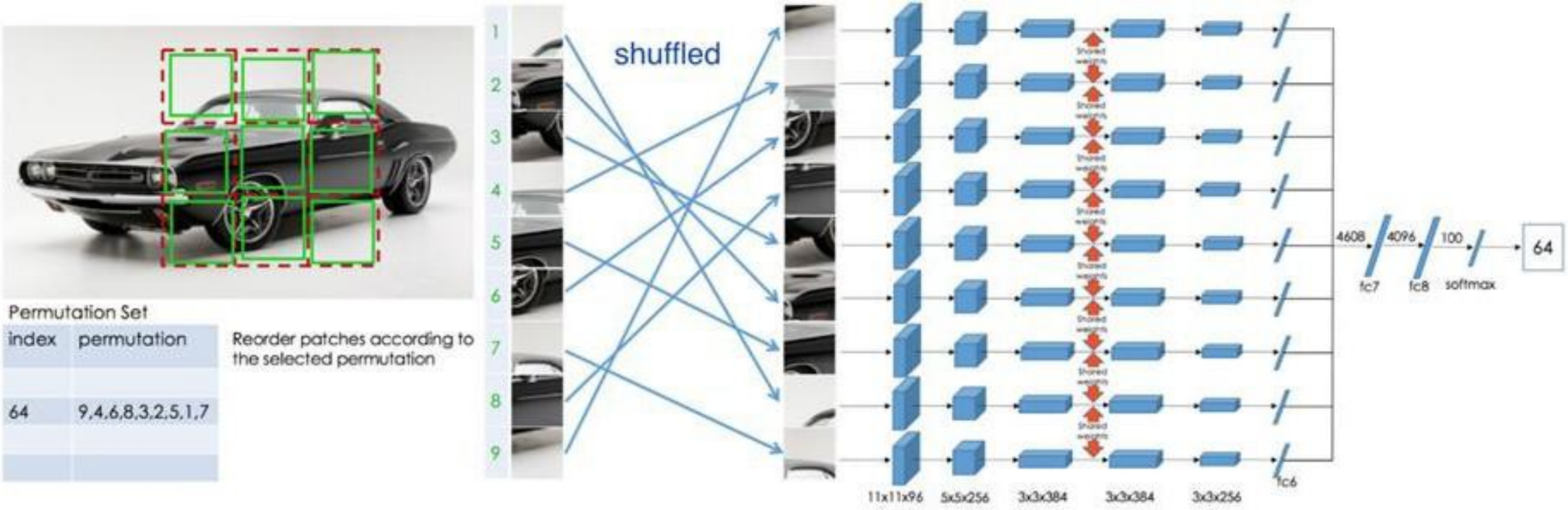
(b) **Attention maps of our self-supervised model**

Pretext task: predict relative patch locations





Pretext task: solving “jigsaw puzzles”



Transfer learned features to supervised learning

Table 1: Results on PASCAL VOC 2007 Detection and Classification. The results of the other methods are taken from Pathak *et al.* [30].

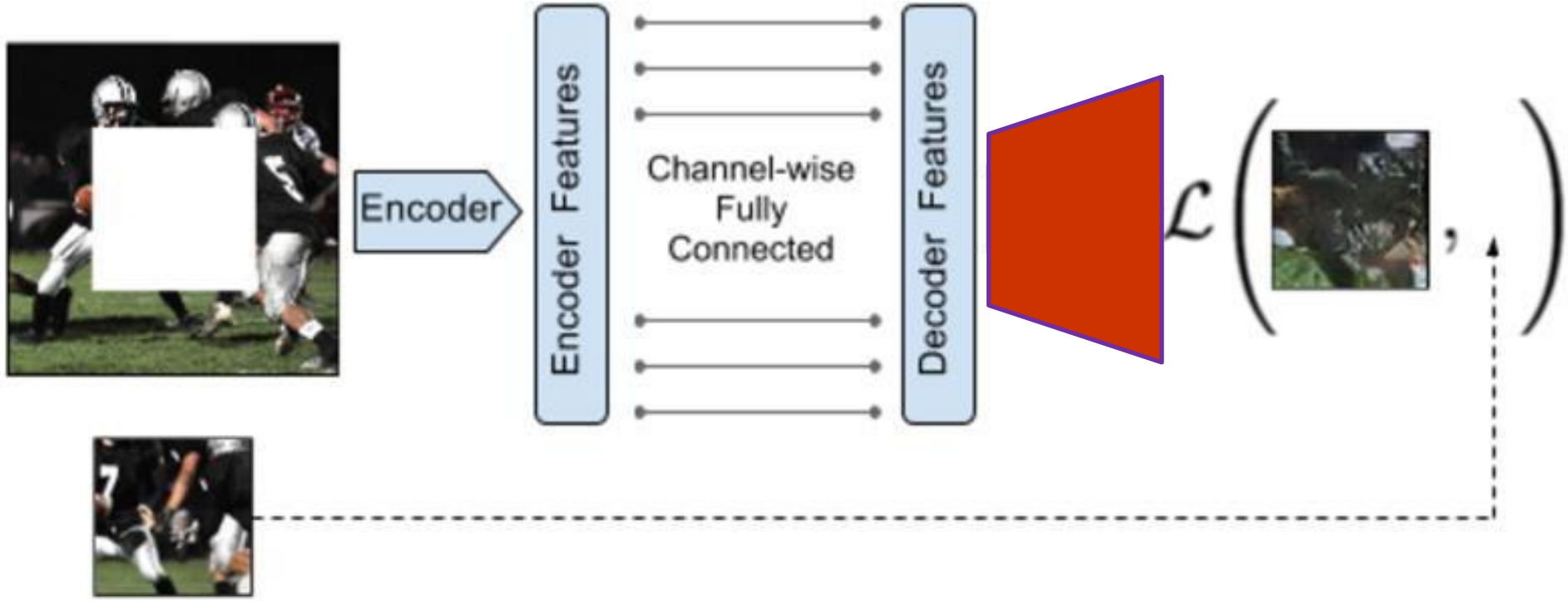
Method	Pretraining time	Supervision	Classification	Detection	Segmentation
Krizhevsky*et al.* [25]	3 days	1000 class labels	**78.2%**	**56.8%**	**48.0%**
Wang and Gupta[39]	1 week	motion	58.4%	44.0%	-
Doersch *et al.* [10]	4 weeks	context	55.3%	46.6%	-
Pathak *et al.* [30]	14 hours	context	56.5%	44.5%	29.7%
Ours	2.5 days	context	**67.6%**	**53.2%**	**37.6%**

- “Ours” is feature learned from solving image Jigsaw puzzles (Noroozi & Favaro, 2016). Doersch et al. is the method with relative patch location

Pretext task: predict missing pixels (inpainting)

Context Encoders: Feature Learning by Inpainting (Pathak et al., 2016)

Learning to inpaint by reconstruction



Learning to reconstruct the missing pixels

Inpainting evaluation

Input (context)

reconstruction

Learning to inpaint by reconstruction

- Loss = reconstruction + adversarial learning
- $L(x) = L_{recon}(x) + L_{adv}(x)$
- $L_{recon}(x) = \left\| M * \left(x - F_\theta\left((1 - M) * x \right) \right) \right\|_2^2$
  - M: mask, 0 indicates not masked; 1 indicates masked
  - F_θ: encoder
- $L_{adv}(x)$: adversarial loss (recall GAN). Loss between “real”images and inpainted images.

Inpainting evaluation

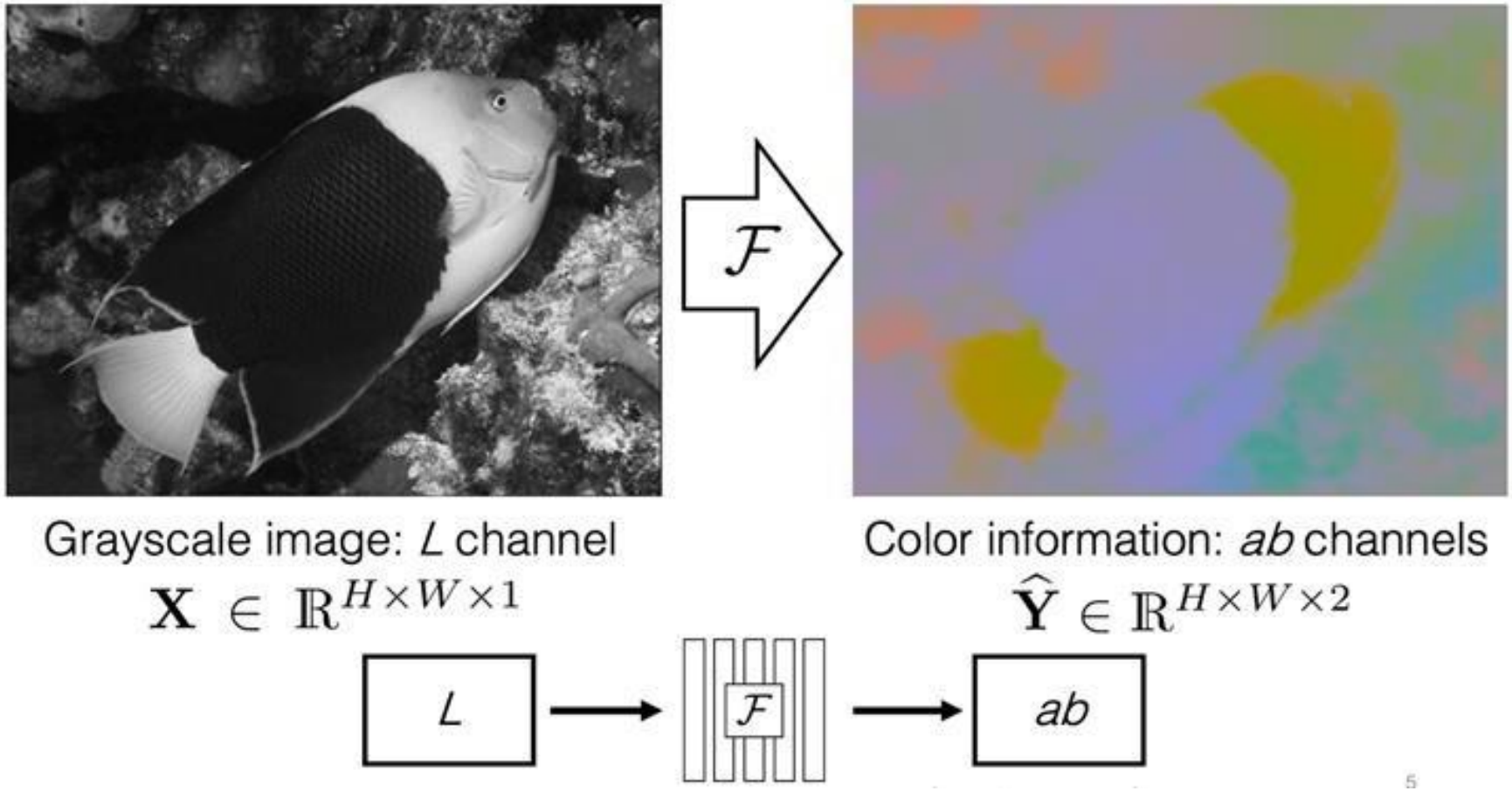
Input (context) reconstruction adversarial recon + adv

Transfer learned features to supervised learning

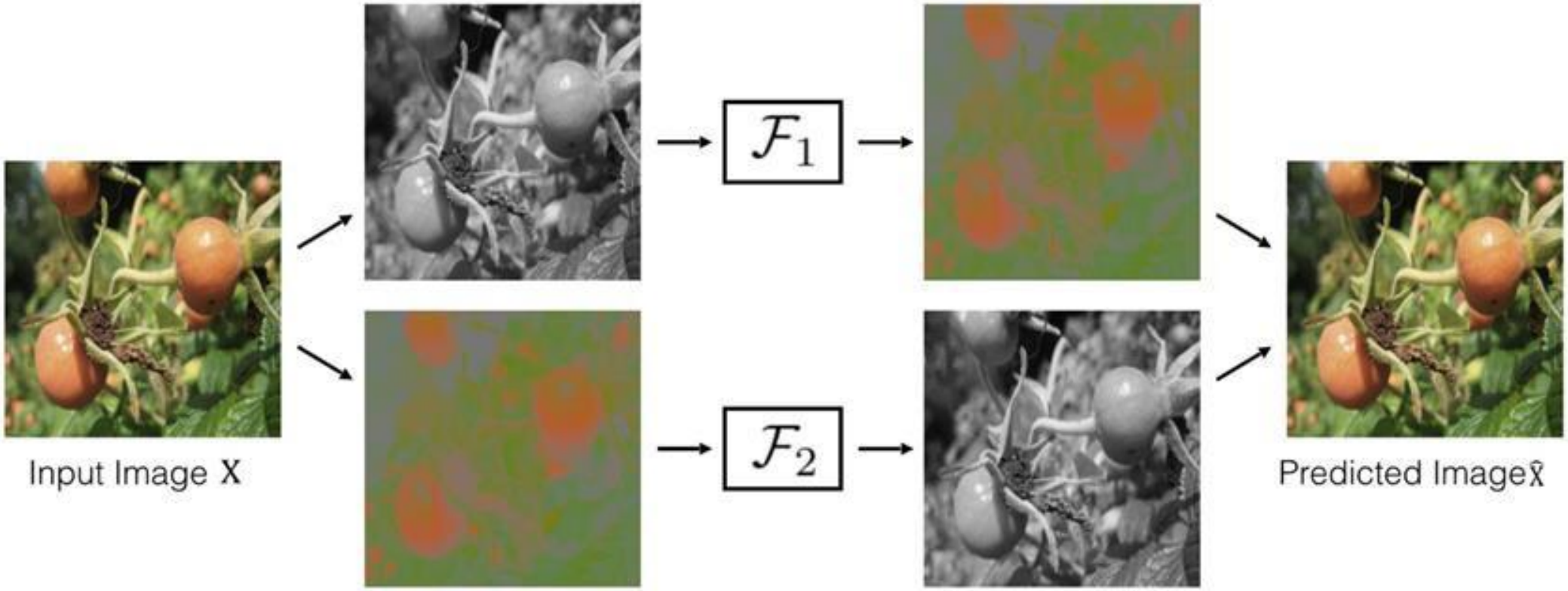
Pretraining Method	Supervision	Pretraining time	Classification	Detection	Segmentation
ImageNet [26]	1000 class labels	3 days	78.2%	56.8%	48.0%
Random Gaussian	initialization	< 1 minute	53.3%	43.4%	19.8%
Autoencoder	-	14 hours	53.8%	41.9%	25.2%
Agrawal *et al.* [1]	egomotion	10 hours	52.9%	41.8%	-
Wang *et al.* [39]	motion	1 week	58.7%	47.4%	-
Doersch *et al.* [7]	relative context	4 weeks	55.3%	46.6%	-
Ours	context	14 hours	56.5%	44.5%	30.0%

- Self-supervised learning on ImageNet training set, transfer to classification (Pascal VOC 2007), detection (Pascal VOC 2007), and semantic segmentation (Pascal VOC 2012)

Pretext task: image coloring

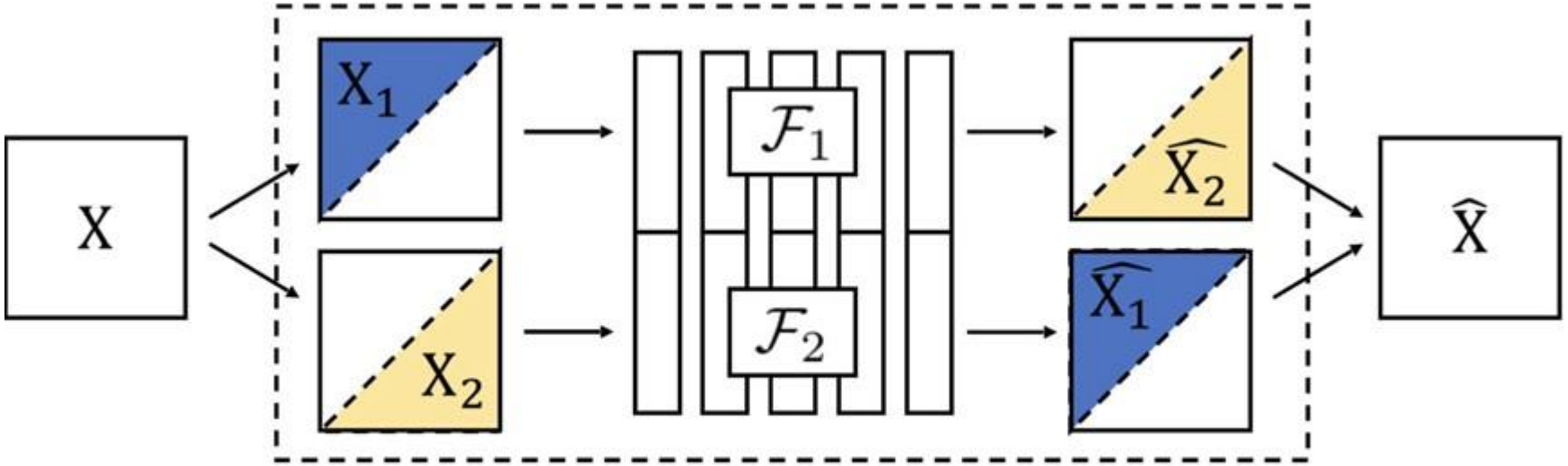


Learning features from colorization: Split-brain Autoencoder



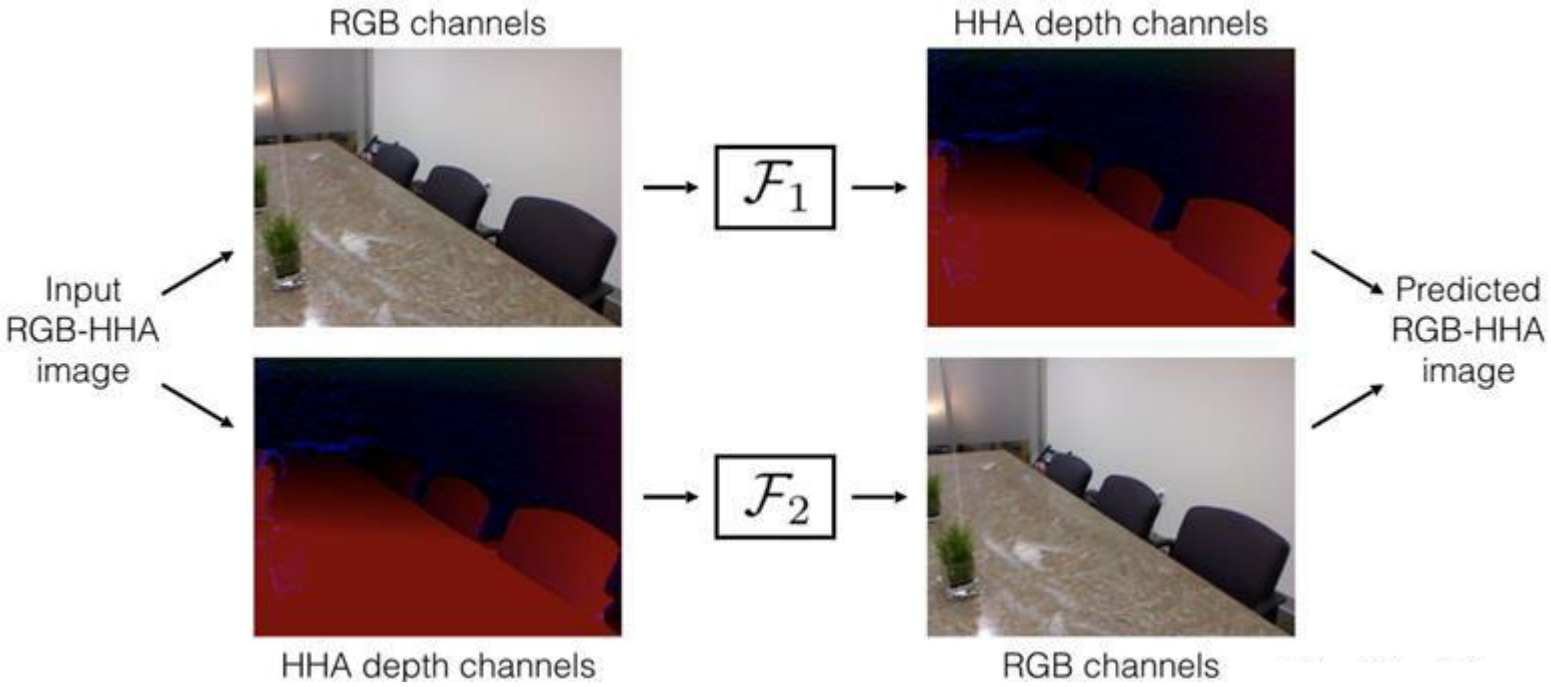
Learning features from colorization: Split-brain Autoencoder

- Idea: cross-channel predictions



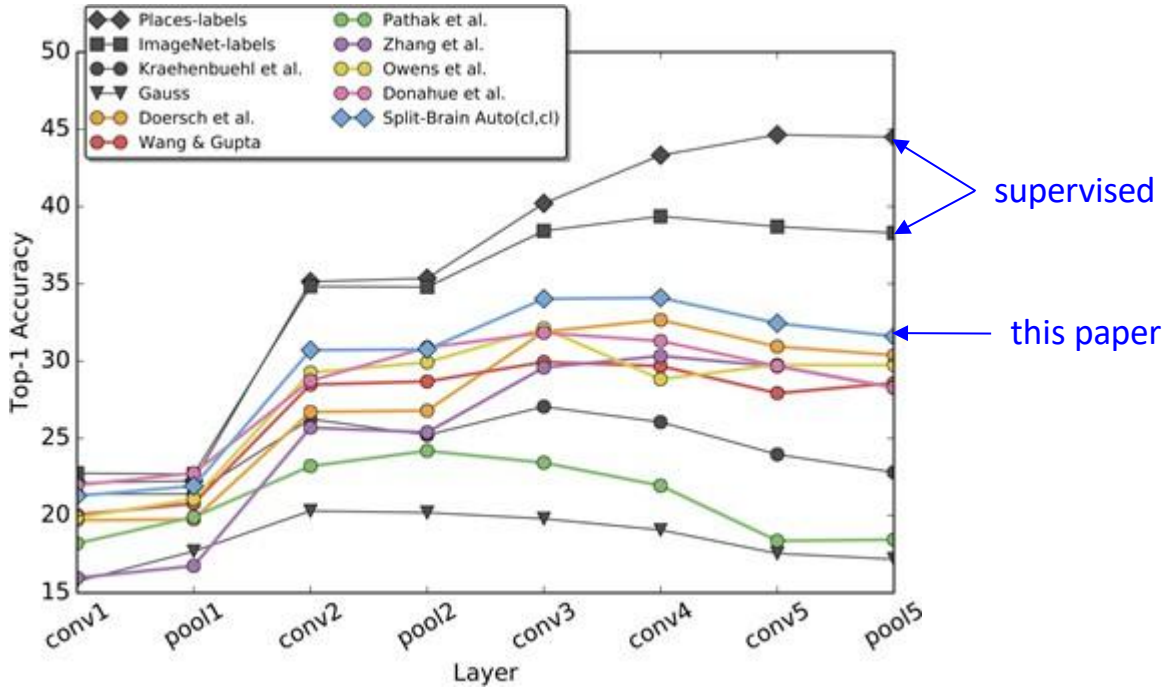
Split-Brain Autoencoder

Learning features from colorization: Split-brain Autoencoder



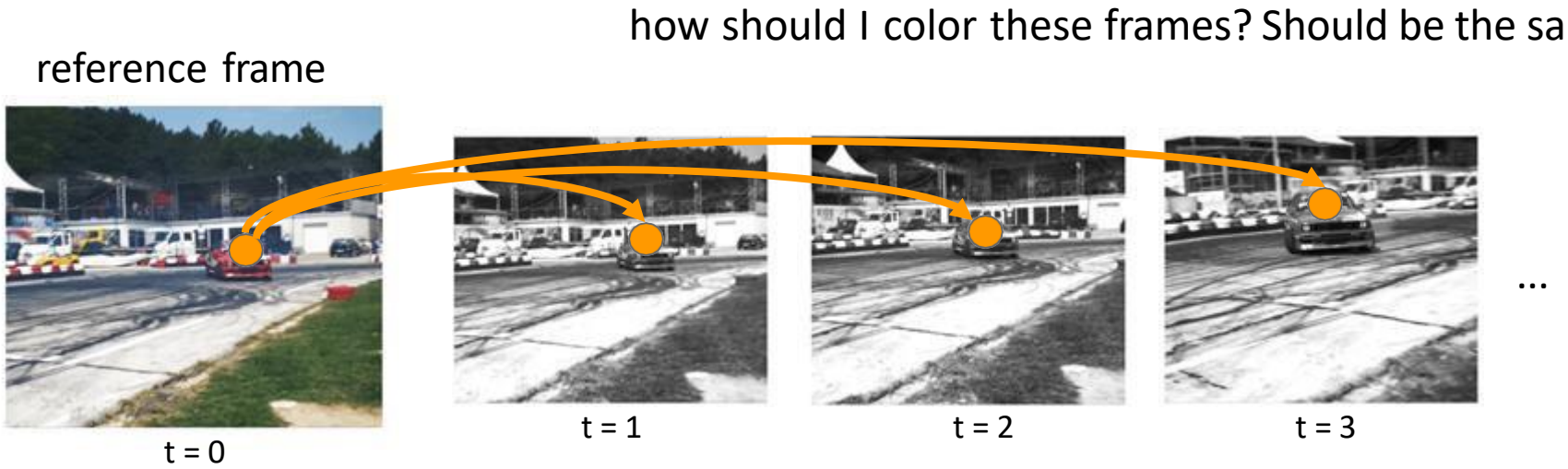
Transfer learned features to supervised learning

- Self-supervised learning on ImageNet (entire training set).
- Use concatenated features from F1 and F2
- Labeled data is from the Places (Zhou 2016).



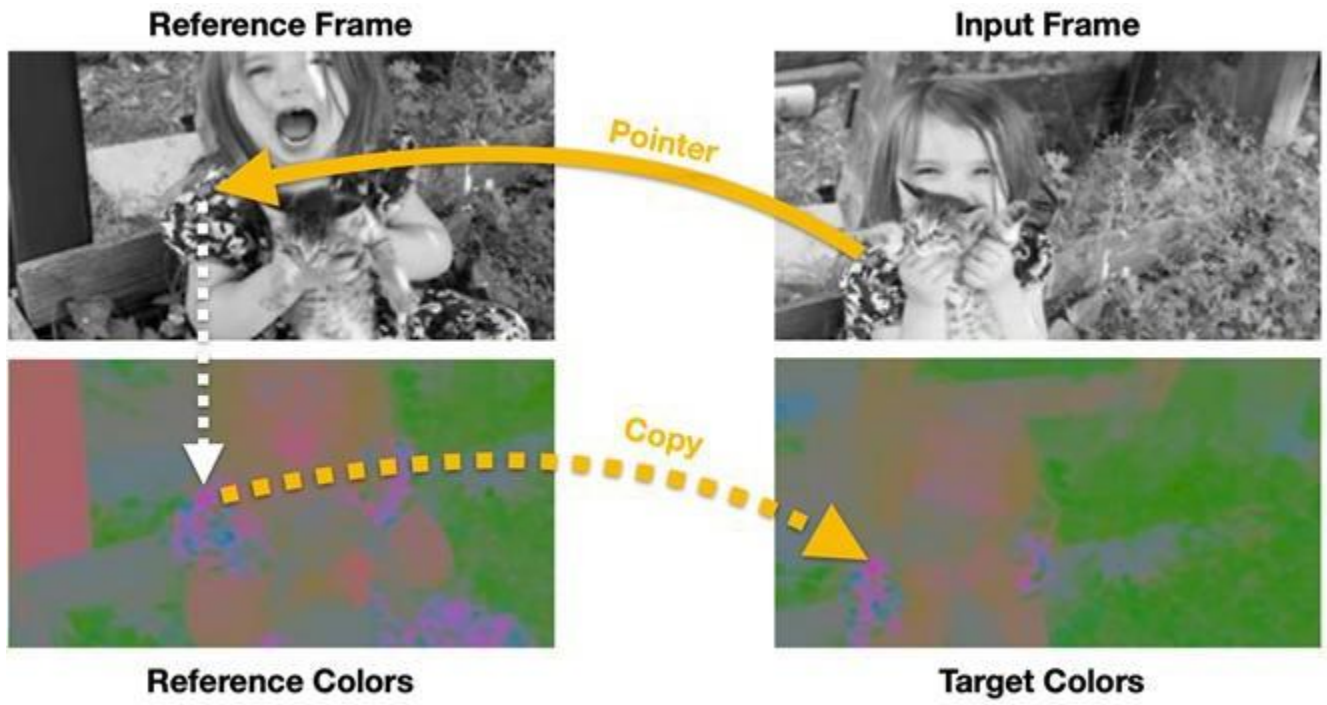
Pretext task: video coloring

- Idea: model the temporal coherence of colors in videos



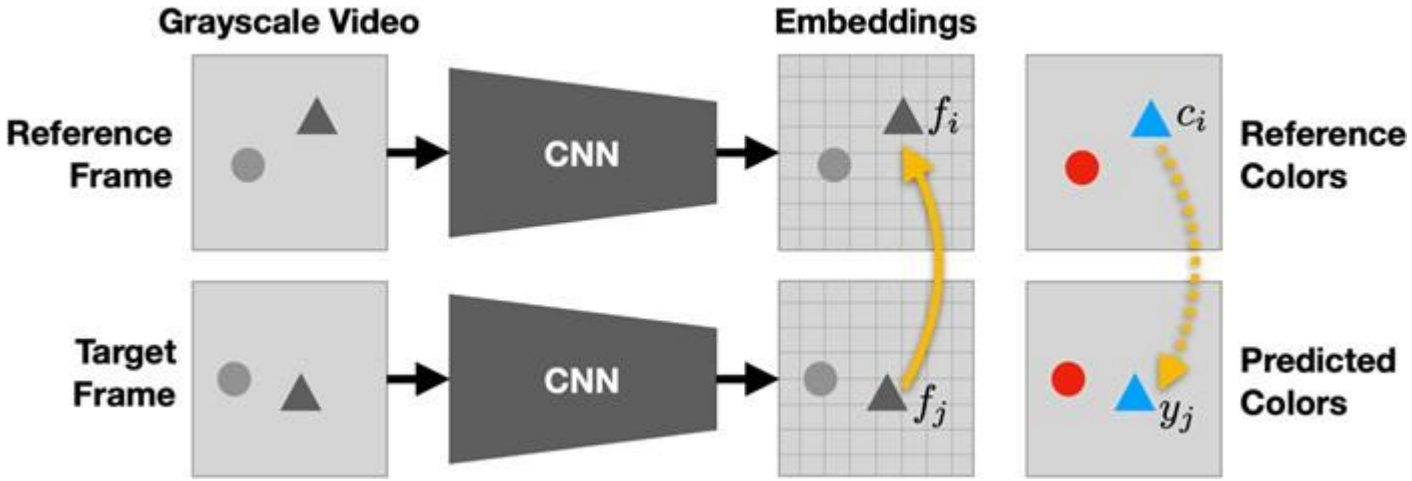
- Hypothesis: learning to color video frames should allow model to learn to track regions or objects without labels!

Learning to color videos



- Learning objective:
- Establish mappings between reference and target frames in a learned feature space.
- Use the mapping as “pointers” to copy the correct color (LAB).

Learning to color videos



- Attention map on the reference frame: $A_{ij} = \frac{\exp(f_i^T f_j)}{\sum_k \exp(f_k^T f_j)}$
- predicted color = weighted sum of the reference color, $y_j = \sum_i A_{ij} c_i$
- loss between predicted color and ground truth color: $\min_\theta \sum_j L(y_j, c_j)$

Colorizing videos (qualitative)

reference frame

target frames (gray)

predicted color

Colorizing videos (qualitative)

reference frame

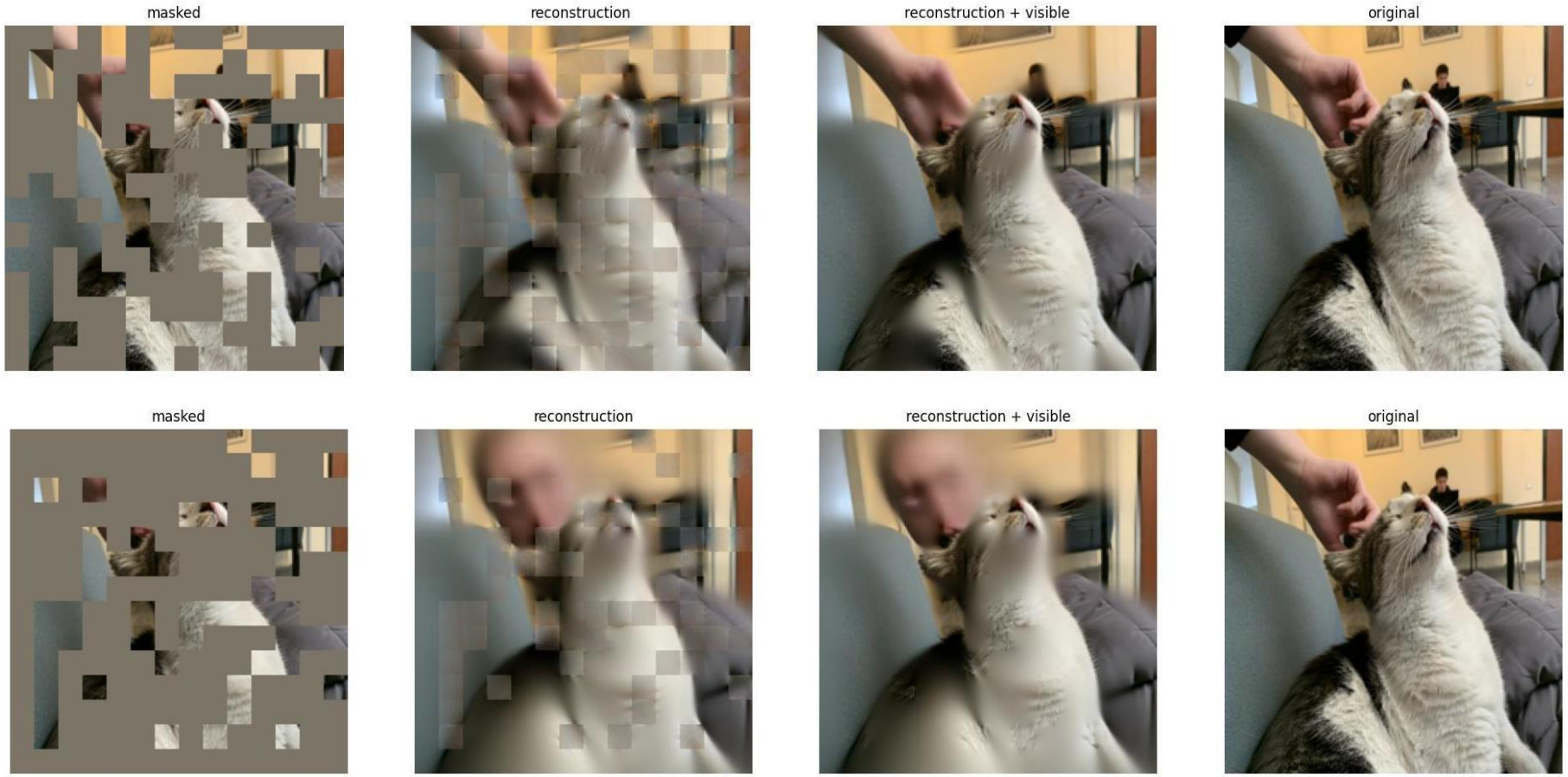
target frames (gray)

predicted color

Masked Auto Encoders (MAE) Reconstruction with a larger masked portion

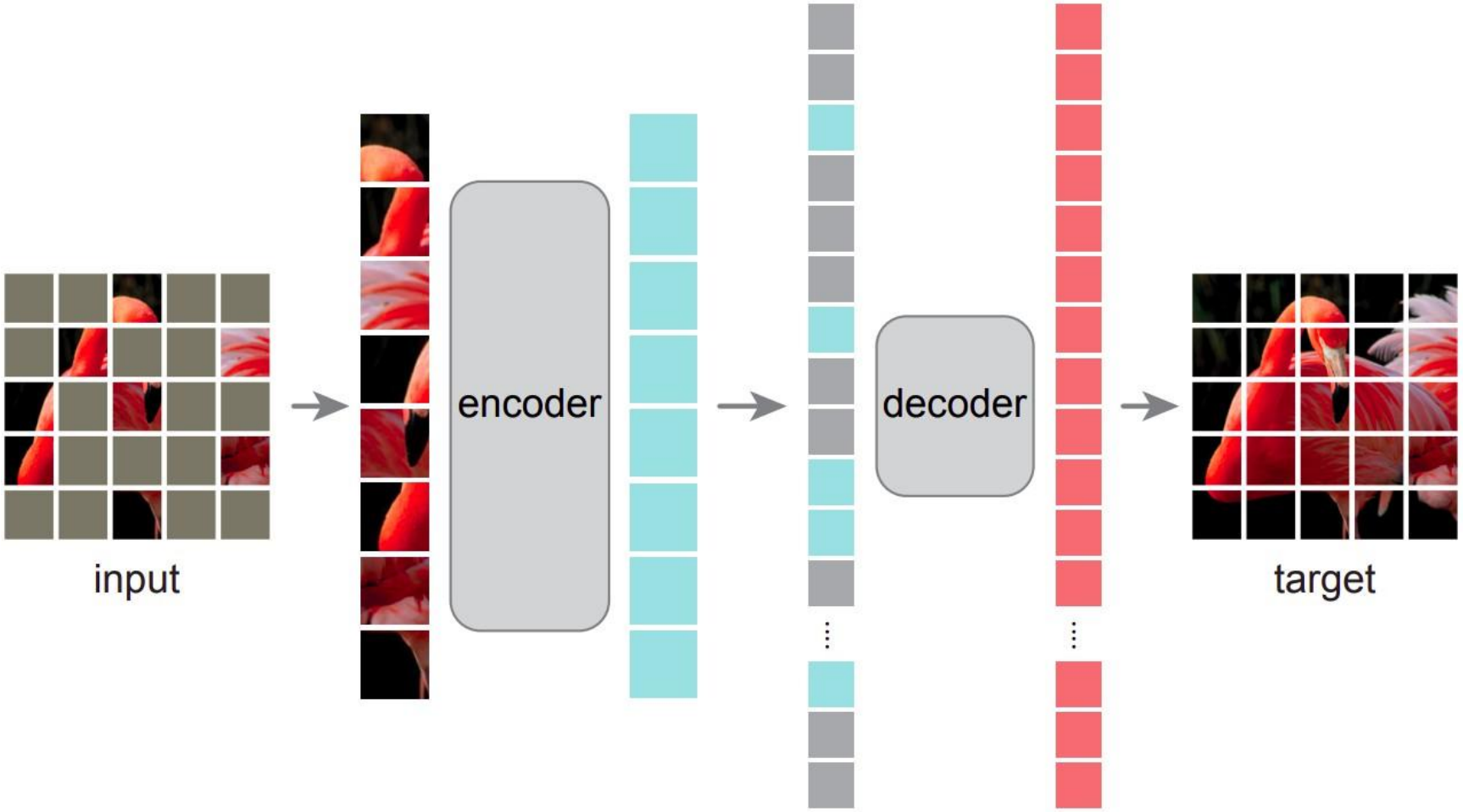
50% masking ratio

75% masking ratio



He et al., 2021 Masked Autoencoders Are Scalable Vision Learners

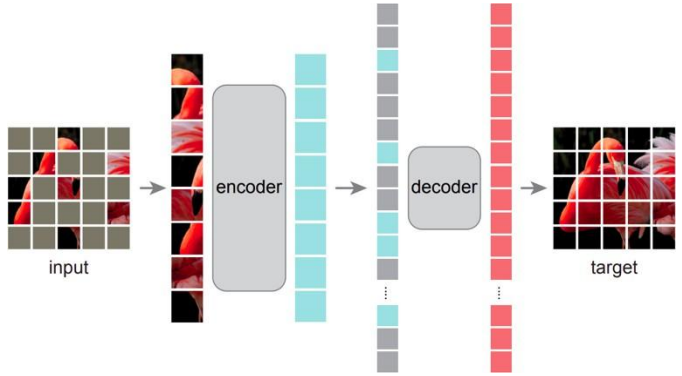
Masked Auto Encoders (MAE)



Masking Methos

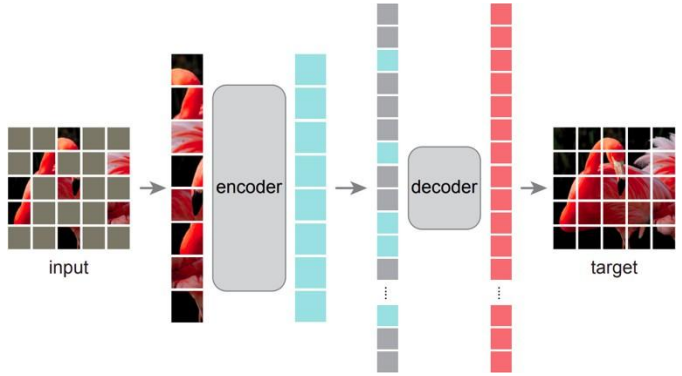
- Similar to the original ViT, divide the input into non-overlapping patches.
- Uniformly sample a very large proportion (75%) of these patches and mask them
- Masking a high ratio makes predicting the task challenging and meaningful.
- Also, not using mask tokens and picking a high sampling ratio (masking most of the image, e.g., 75%, and sampling a small visible portion, e.g., 25%, to feed into the encoder) enables the encoder to be very large.

MAE Encoder



- The encoder only operates on unmasked patches (25%)
- Embeds the patches by linear projection and add positional embeddings
- Uses transformer blocks
- Since the input patches are a small part of the input, the encoder is chosen to be very large.
- Encoder has over 9 times computations per token vs decoder

MAE Decoder



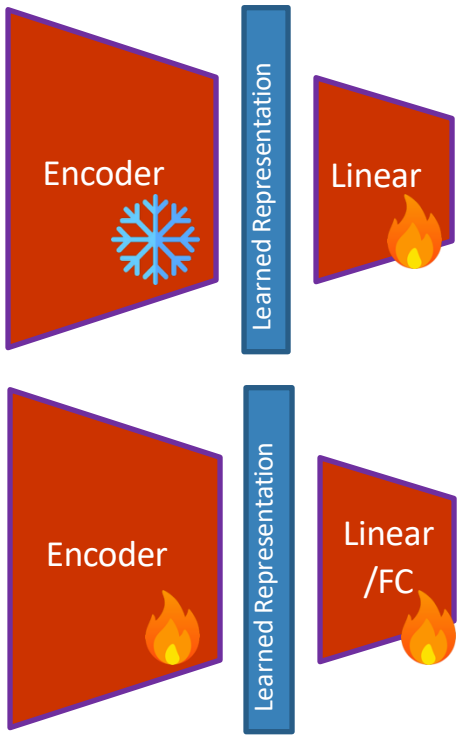
- Merges the encoder outputs with the shared mask tokens in previously masked places,adding positional encodings to them.
- Uses transformer blocks, followed by a linear projection for finalizing pixel reconstruction.
- Is solely responsible for reconstruction, meaning it is not used post-training.
- Hence, it is independent of the encoder design, making it flexible (unlike traditional AEs or UNet).
- This is an asymmetrical autoencoder design.

Reconstruction

- The MSE (mean squared error loss) in the pixel space between the input image and the reconstructed image is adopted.
- Loss is only computed for masked patches

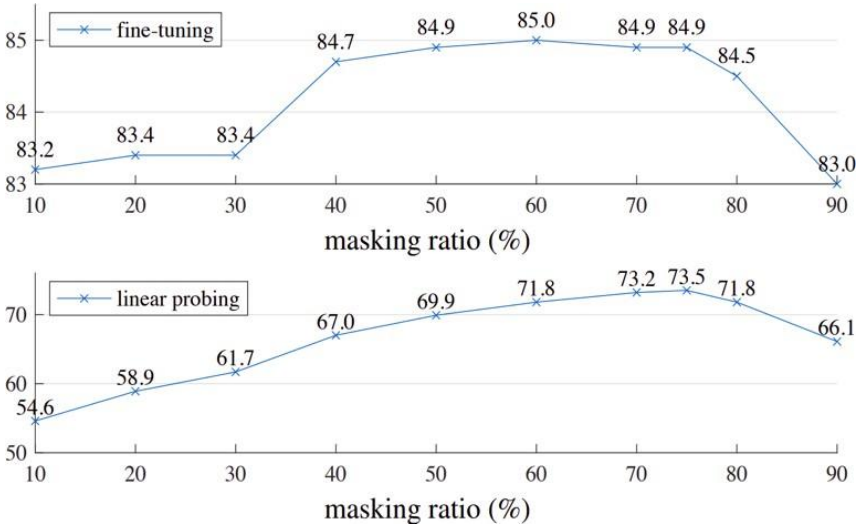
Linear Probing vs Full Fine-tuning

- In linear probing, the pre-trained model is fixed, and only one linear layer is added at the end, to predict the labels (or produce the output). This method is used to assess the quality of representations from a pre-trained feature extraction model.
- In fine-tuning, pre-trained model is further trained (not fixed), and one or more layers, possibly with non-linearities are added.
- linear probing: provides a measure of representation quality of a pre-training in restricted conditions
- fine-tuning: exploits models near-true potential to adopt for new tasks



Ablation Studies

- So many modeling/hyperparameter choices:
  - Masking ratio
  - Decoder depth
  - Decoder width
  - Mask token (used or not in encoder)
  - Reconstruction target
  - Data augmentation
  - Mask sampling method
  - Training schedule



case	ratio	ft	lin
random	75	**84.9**	**73.5**
block	50	83.9	72.3
block	75	82.8	63.9
grid	75	84.0	66.0

random 75% block 50% grid 75%

Summary: pretext tasks from image transformations

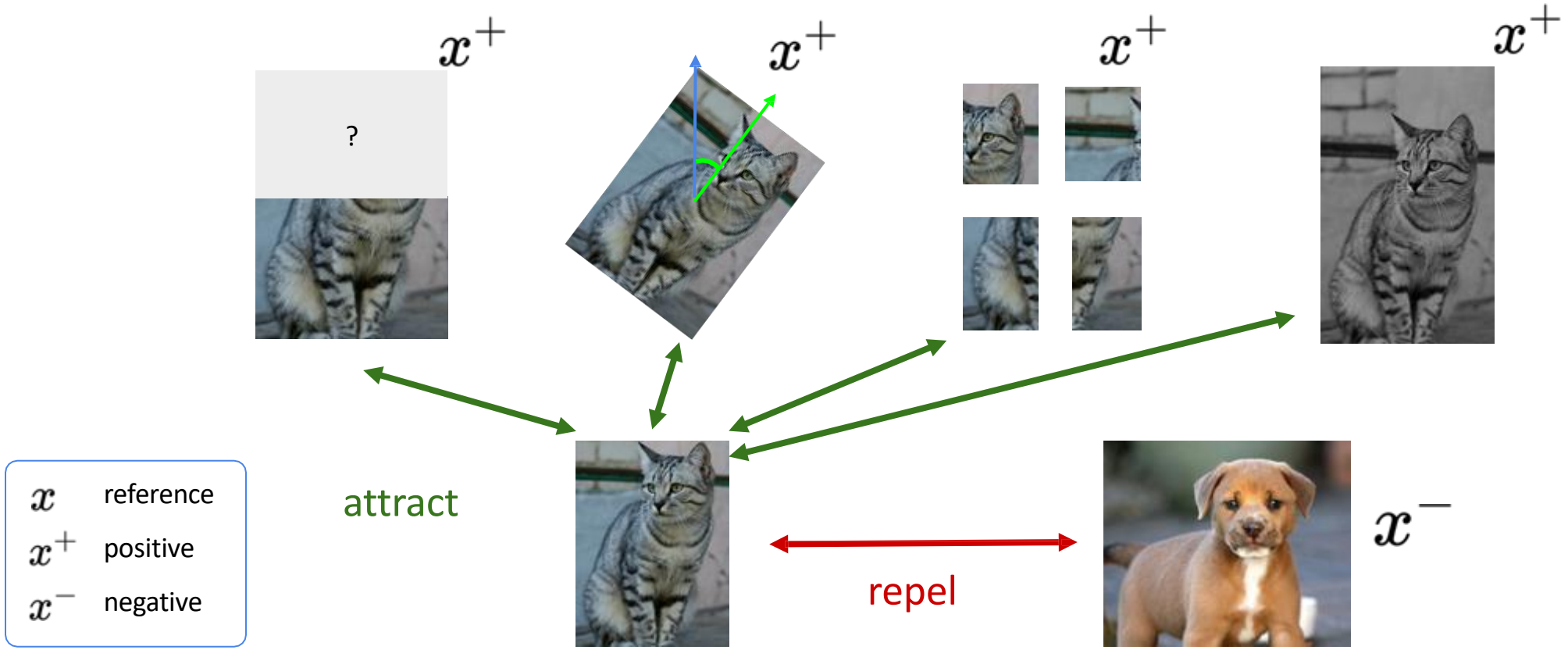
- Pretext tasks focus on “visual common sense”, e.g., predict rotations, inpainting, rearrangement, and colorization.
- The models are forced learn good features about natural images, e.g., semantic representation of an object category, in order to solve the pretext tasks.
- We often do not care about the performance of these pretext tasks, but rather how useful the learned features are for downstream tasks (classification, detection, segmentation).
- Problems:
  - 1) coming up with individual pretext tasks is tedious, and
  - 2) the learned representations may not be general

Contrastive representation learning

Morden Pretext Tasks Approaches

- Pretext tasks from image transformations
  - Rotation, inpainting, rearrangement, coloring
  - Reconstruction-based learning (MAE)
- Contrastive representation learning
  - Intuition and formulation
  - Instance contrastive learning: SimCLR and MOCO
  - Sequence contrastive learning: CPC
  - Self-Distillation Without Labels, DINO

Contrastive Representation Learning

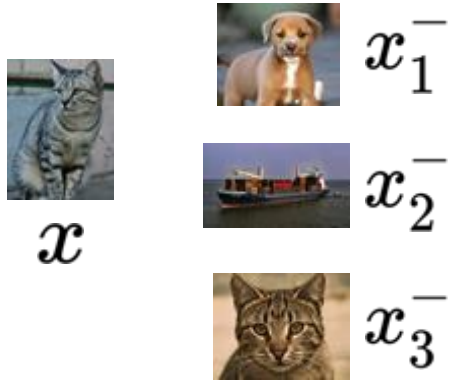


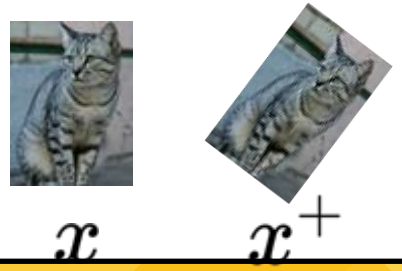
A formulation of contrastive learning

- What we want:

$$score\big(f(x), f(x^+)\big) \gcurlyeq score(f(x), f(x^-))$$

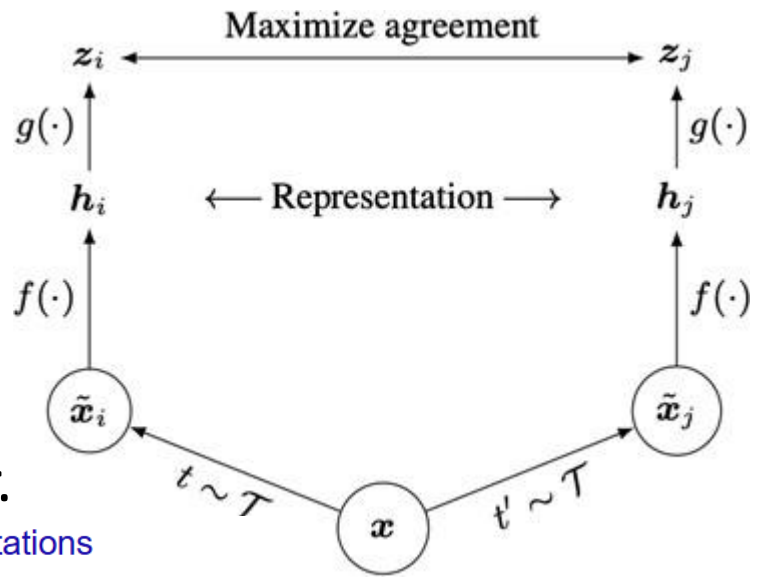
- x: reference sample; x^+ positive sample; x^- negative sample
- Given a chosen score function, we aim to learn an encoder function f that yields high score for positive pairs (x, x+) and low scores for negative pairs (x, x-).
- Loss function given 1 positive sample and $N - 1$ negative samples
- $L = -E_x\left[\log \frac{\exp\left(s\left(f(x), f(x^+)\right)\right)}{\exp\left(s\left(f(x), f(x^+)\right)\right) + \sum_{j=1}^{N-1} \exp\left(s\left(f(x), f\left(x_j^-\right)\right)\right)}\right]$





SimCLR: A Simple Framework for Contrastive Learning

- Cosine similarity as the score function: $s(u, v) = \frac{u^T v}{||u|| \cdot ||v||}$
- Use a projection network g(·) to project features to a space where contrastive learning is applied
- Generate positive samples through data augmentation:
- random cropping, random color distortion, and random blur.



A simple framework for contrastive learning of visual representations

T Chen, S Kornblith, M Norouzi, G Hinton

International conference on machine learning, 2020 • proceedings.mlr.press

Abstract

This paper presents SimCLR: a simple framework for contrastive learning of visual representations. We simplify recently proposed contrastive self-supervised learning algorithms without requiring specialized architectures or a memory bank. In order to understand what enables the contrastive prediction tasks to learn useful representations, we systematically study the major components of our framework. We show that (1) composition of data augmentations plays a critical role in defining effective predictive

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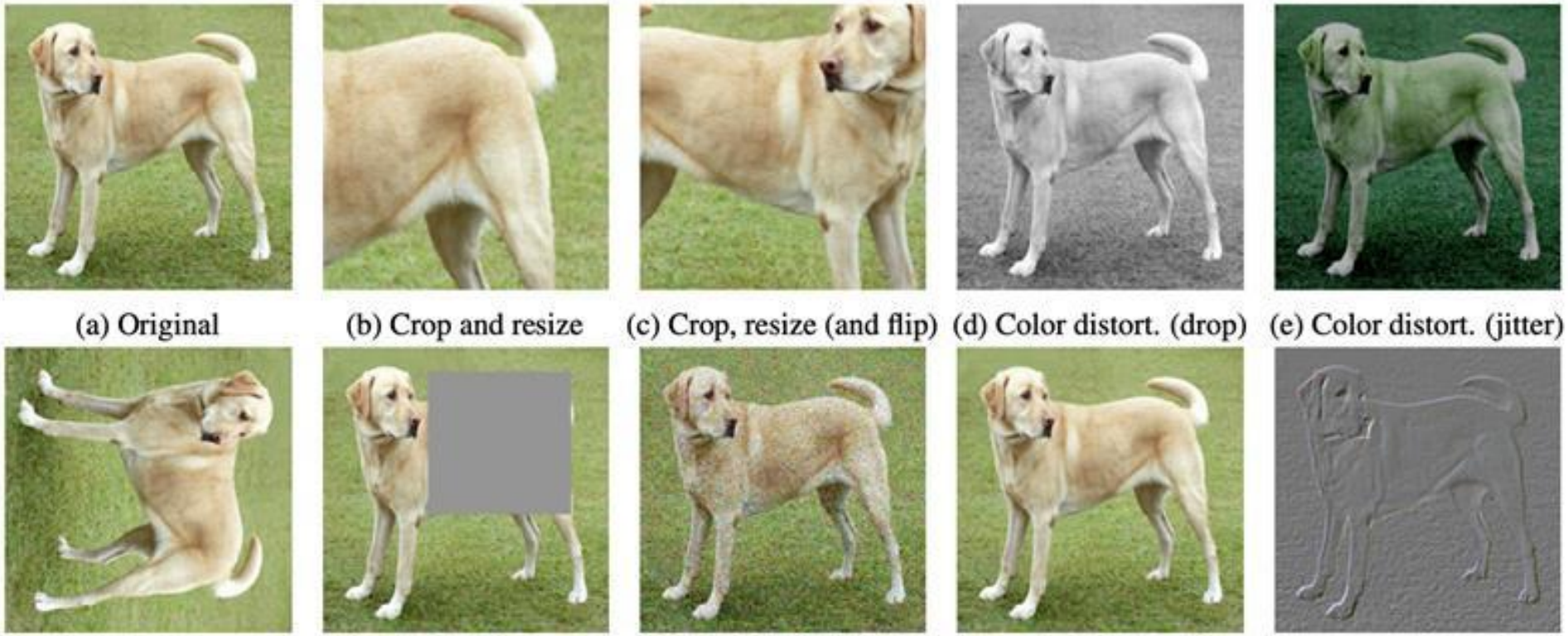
SimCLR: generating positive samples from data augmentation



(a) Original (b) Crop and resize (c) Crop, resize (and flip) (d) Color distort. (drop) (e) Color distort. (jitter)

(f) Rotate $\{90°, 180°, 270°\}$ (g) Cutout (h) Gaussian noise (i) Gaussian blur (j) Sobel filtering

SimCLR

Generate a positive pair by sampling data augmentation functions

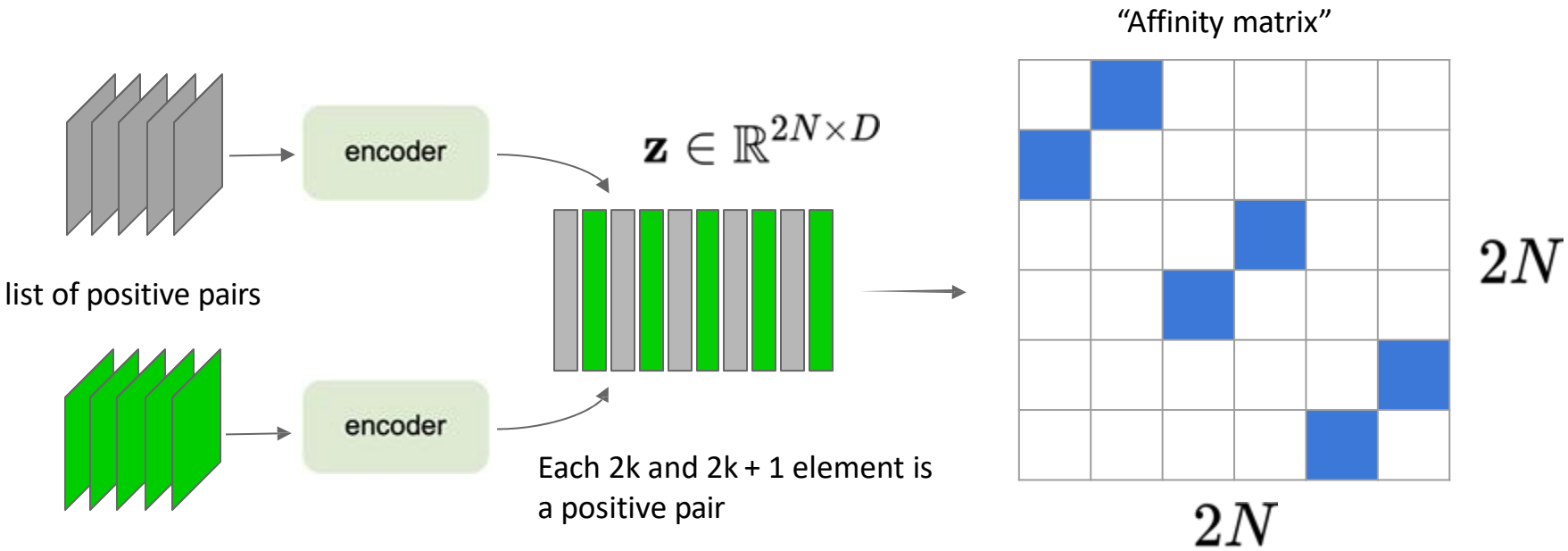
Algorithm 1 SimCLR's main learning algorithm.

input: batch size N, constant τ, structure of $f, g, \mathcal{T}$.
for sampled minibatch $\{\boldsymbol{x}_k\}_{k=1}^{N}$ **do**
  for all $k \in \{1, \ldots, N\}$ **do**
    draw two augmentation functions $t \sim \mathcal{T}, t' \sim \mathcal{T}$
    # the first augmentation
    $\tilde{\boldsymbol{x}}_{2k-1} = t(\boldsymbol{x}_k)$
    $\boldsymbol{h}_{2k-1} = f(\tilde{\boldsymbol{x}}_{2k-1})$ # representation
    $\boldsymbol{z}_{2k-1} = g(\boldsymbol{h}_{2k-1})$ # projection
    # the second augmentation
    $\tilde{\boldsymbol{x}}_{2k} = t'(\boldsymbol{x}_k)$
    $\boldsymbol{h}_{2k} = f(\tilde{\boldsymbol{x}}_{2k})$ # representation
    $\boldsymbol{z}_{2k} = g(\boldsymbol{h}_{2k})$ # projection
  end for
  for all $i \in \{1, \ldots, 2N\}$ and $j \in \{1, \ldots, 2N\}$ **do**
    $s_{i,j} = \boldsymbol{z}_i^\top \boldsymbol{z}_j / (\|\boldsymbol{z}_i\| \|\boldsymbol{z}_j\|)$ # pairwise similarity
  end for
  define $\ell(i, j)$ **as** $\ell(i,j) = -\log \frac{\exp(s_{i,j}/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(s_{i,k}/\tau)}$
  $\mathcal{L} = \frac{1}{2N} \sum_{k=1}^{N} [\ell(2k-1, 2k) + \ell(2k, 2k-1)]$
  update networks f and g to minimize $\mathcal{L}$
end for
return encoder network $f(\cdot)$, and throw away $g(\cdot)$

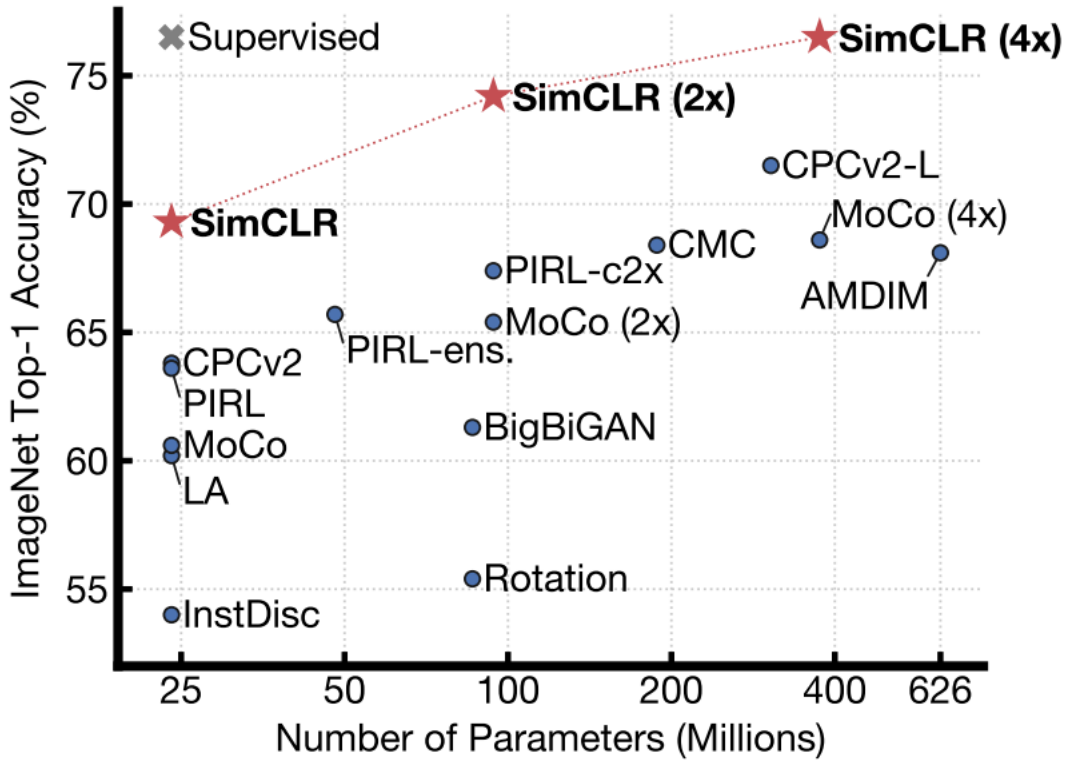
Use all non-positive samples in the batch as x -

SimCLR: mini-batch training

$$s_{i,j} = \frac{z_i^T z_j}{||z_i||\,||z_j||}$$



Training linear classifier on SimCLR features



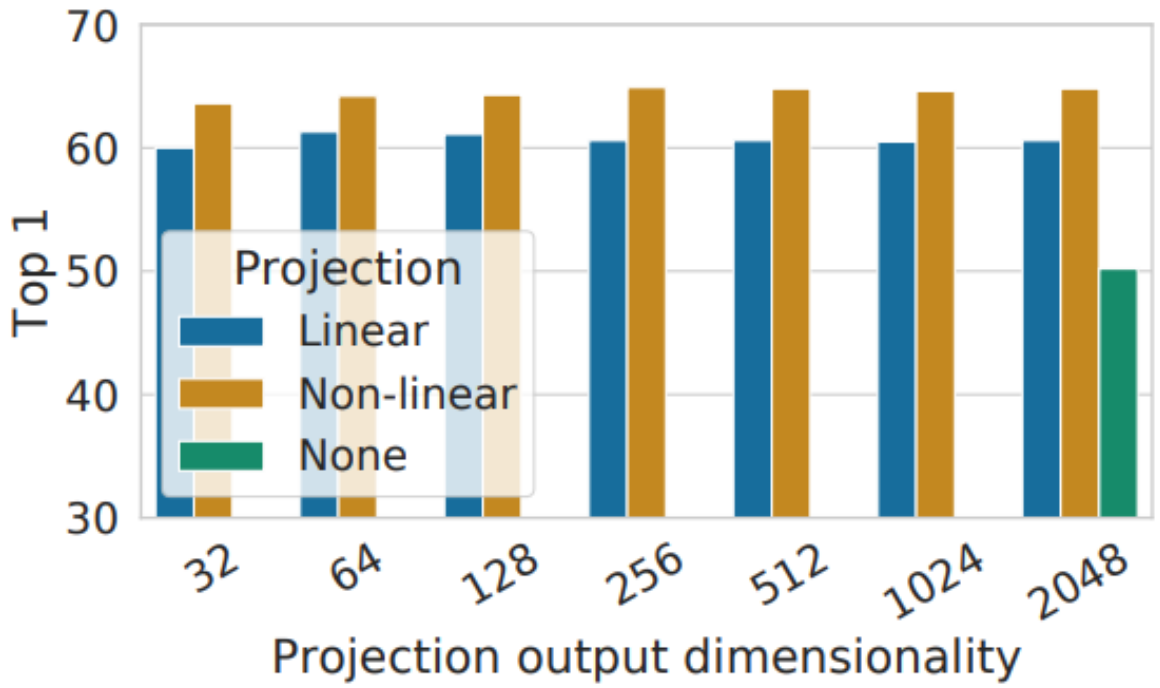
- Train feature encoder on ImageNet (entire training set) using SimCLR.
- Freeze feature encoder, train a linear classifier on top with labeled data.

Semi-supervised learning on SimCLR features

Method	Architecture	Param (M)	Top 1	Top 5
Methods using ResNet-50:				
Local Agg.	ResNet-50	24	60.2	-
MoCo	ResNet-50	24	60.6	-
PIRL	ResNet-50	24	63.6	-
CPC v2	ResNet-50	24	63.8	85.3
SimCLR (ours)	ResNet-50	24	**69.3**	**89.0**
Methods using other architectures:				
Rotation	RevNet-50 (4×)	86	55.4	-
BigBiGAN	RevNet-50 (4×)	86	61.3	81.9
AMDIM	Custom-ResNet	626	68.1	-
CMC	ResNet-50 (2×)	188	68.4	88.2
MoCo	ResNet-50 (4×)	375	68.6	-
CPC v2	ResNet-161 (∗)	305	71.5	90.1
SimCLR (ours)	ResNet-50 (2×)	94	74.2	92.0
SimCLR (ours)	ResNet-50 (4×)	375	**76.5**	**93.2**

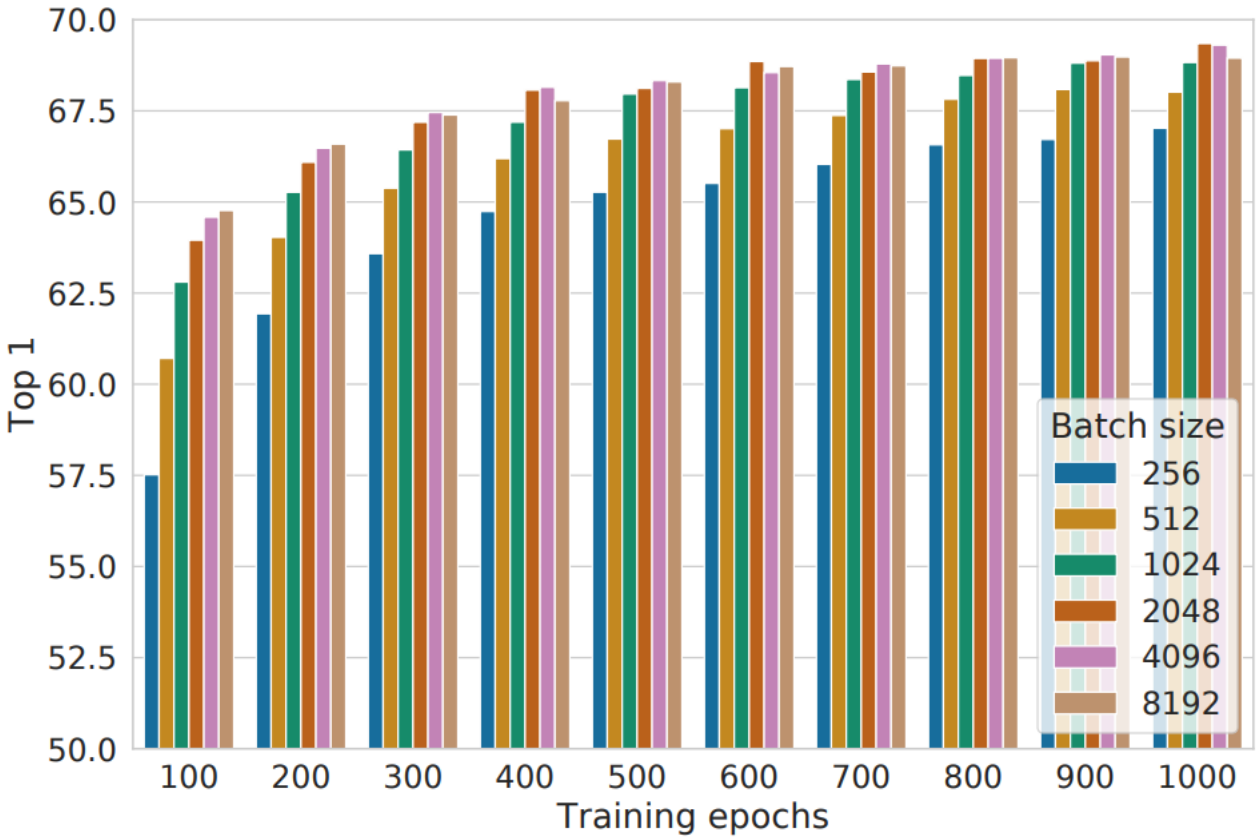
- ImageNet accuracies of linear classifiers trained on representations learned with different self-supervised methods.
- Finetune the encoder with 1% / 10% of labeled data on ImageNet.

SimCLR design



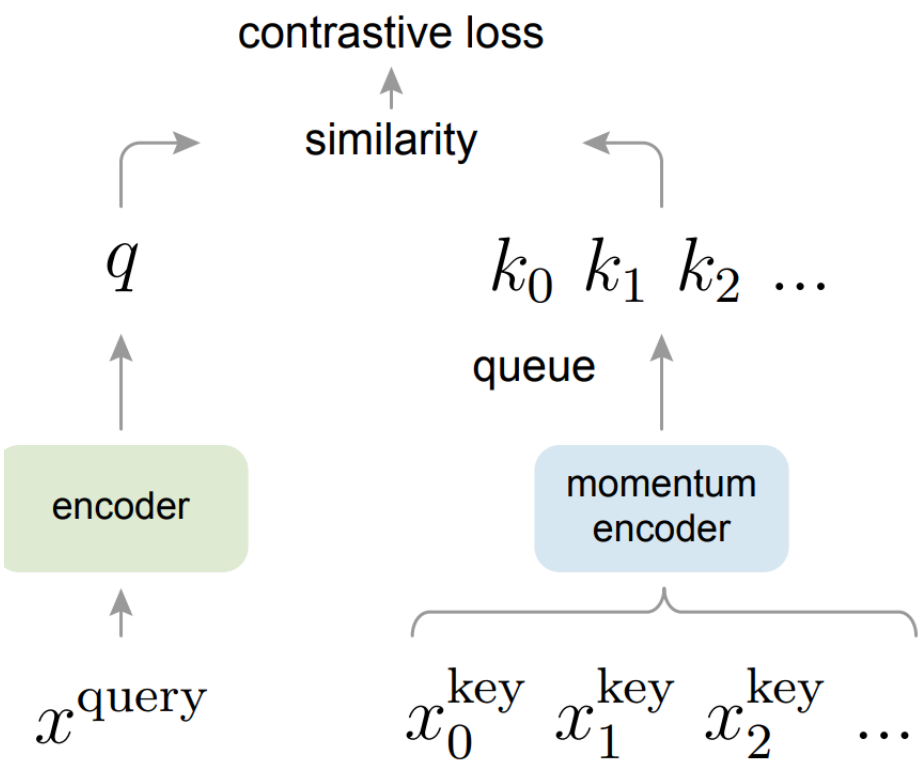
- Linear / non-linear projection heads improve representation learning.
- A possible explanation:
  - contrastive learning objective may discard
  - useful information for downstream tasks
  - representation space z is trained to be invariant to data transformation.
  - by leveraging the projection head g(·), more information can be preserved in the h representation space

SimCLRdesign choices: large batch size



- Large training batch size is crucial for SimCLR!
- Large batch size causes large memory footprint during backpropagation:
- Requires distributed training on TPUs (ImageNet experiments)

Momentum Contrastive Learning (MoCo)



- Differences to SimCLR:
- Keep a running queue of keys (negative samples).
- Compute gradients and update the encoder only through the queries.
- Decouple min-batch size with the number of keys: can support a large number of negative samples.
- The key encoder is slowly progressing through the momentum update rules:

$$\theta_k \leftarrow m\theta_k + (1-m)\theta_q$$

Momentum Contrastive Learning (MoCo)

Algorithm 1 Pseudocode of MoCo in a PyTorch-like style.

```
# f_q, f_k: encoder networks for query and key
# queue: dictionary as a queue of K keys (CxK)
# m: momentum
# t: temperature

f_k.params = f_q.params # initialize
for x in loader: # load a minibatch x with N samples
  x_q = aug(x) # a randomly augmented version
  x_k = aug(x) # another randomly augmented version

  q = f_q.forward(x_q) # queries: NxC
  k = f_k.forward(x_k) # keys: NxC
  k = k.detach() # no gradient to keys

  # positive logits: Nx1
  l_pos = bmm(q.view(N,1,C), k.view(N,C,1))

  # negative logits: NxK
  l_neg = mm(q.view(N,C), queue.view(C,K))

  # logits: Nx(1+K)
  logits = cat([l_pos, l_neg], dim=1)

  # contrastive loss, Eqn.(1)
  labels = zeros(N) # positives are the 0-th
  loss = CrossEntropyLoss(logits/t, labels)

  # SGD update: query network
  loss.backward()
  update(f_q.params)

  # momentum update: key network
  f_k.params = m*f_k.params+(1-m)*f_q.params

  # update dictionary
  enqueue(queue, k) # enqueue the current minibatch
  dequeue(queue) # dequeue the earliest minibatch
```

bmm: batch matrix multiplication; mm: matrix multiplication; cat: concatenation.

Momentum contrast for **unsupervised visual representation learning**

K He, H Fan, Y Wu, S Xie… - … on computer **vision** and …, 2020 - openaccess.thecvf.com

… **Momentum Contrast** (MoCo) as a way of building large and consistent dictionaries for **unsupervised learning** … key encoder, implemented as a **momentum**-based moving average of the …

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Generate a positive pair by sampling data augmentation functions

Use the running queue of keys as the negative samples

Contrastive loss

Update through momentum

Update the queue

MoCo V2

- A hybrid of ideas from SimCLR and MoCo:
- From SimCLR: non-linear projection head and strong data augmentation.
- From MoCo: momentum-updated queues that allow training on a large number of negative samples (no TPU required!).

Improved baselines with momentum contrastive learning
X Chen, H Fan, R Girshick, K He - arXiv preprint arXiv:2003.04297, 2020 - arxiv.org
Contrastive unsupervised learning has recently shown encouraging progress, eg, in Momentum Contrast (MoCo) and SimCLR. In this note, we verify the effectiveness of two of SimCLR's design improvements by implementing them in the MoCo framework. With simple modifications to MoCo---namely, using an MLP projection head and more data augmentation---we establish stronger baselines that outperform SimCLR and do not require large training batches. We hope this will make state-of-the-art unsupervised learning …
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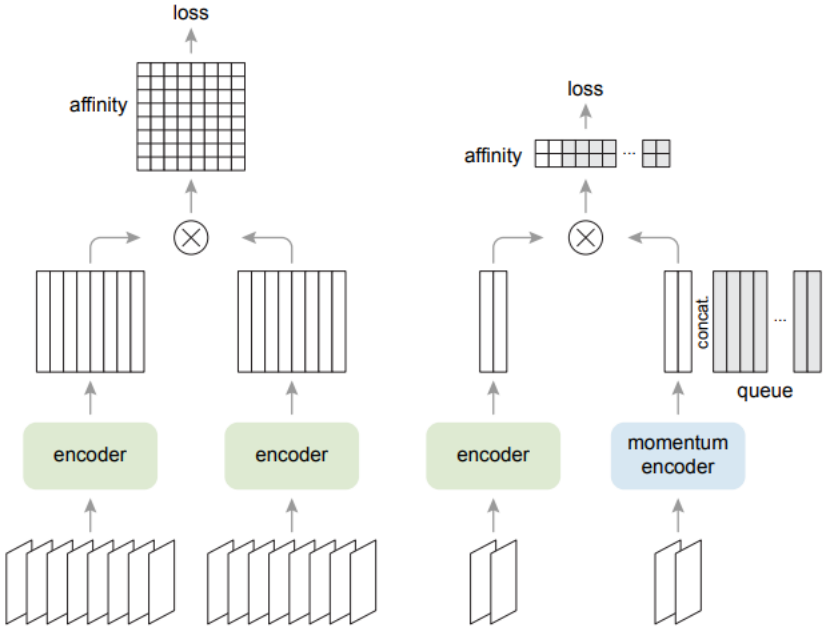


(a) end-to-end (b) Momentum Contrast

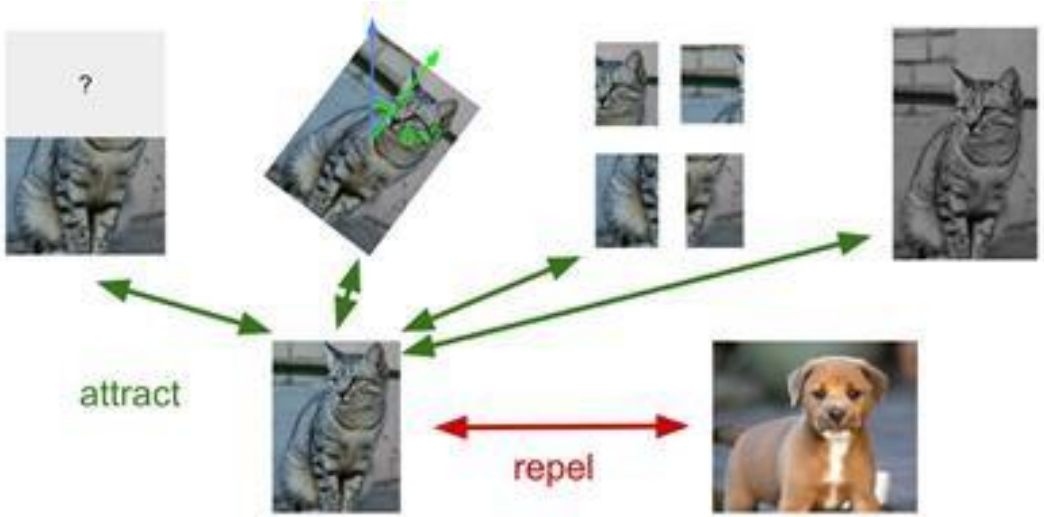
MoCo vs. SimCLR vs. MoCo V2

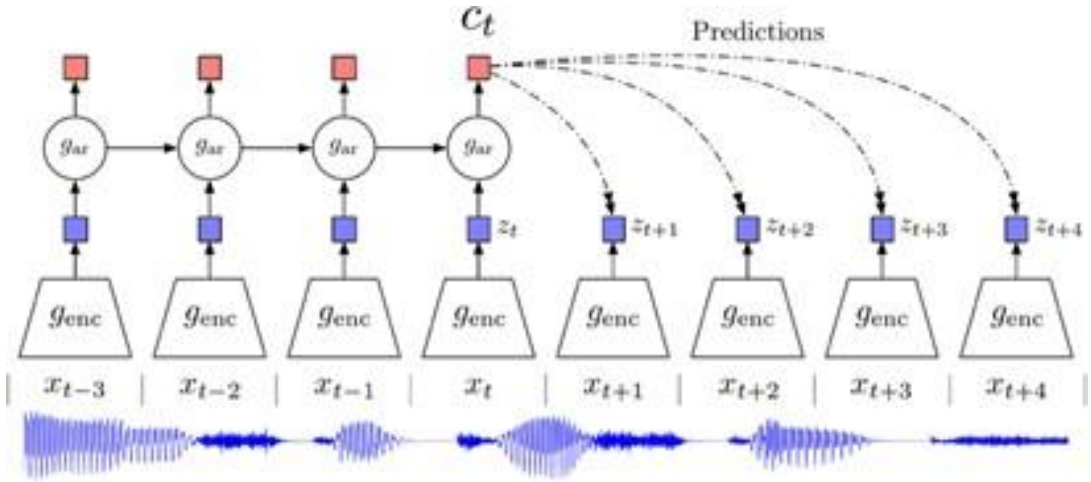
case	unsup. pre-train MLP	aug+	cos	epochs	batch	ImageNet acc.
MoCo v1 [6]				200	256	60.6
SimCLR [2]	✓	✓	✓	200	256	61.9
SimCLR [2]	✓	✓	✓	200	8192	66.6
MoCo v2	✓	✓	✓	200	256	**67.5**
*results of **longer** unsupervised training follow:*						
SimCLR [2]	✓	✓	✓	1000	4096	69.3
MoCo v2	✓	✓	✓	800	256	**71.1**

Table 2. **MoCo *vs.* SimCLR**: ImageNet linear classifier accuracy (**ResNet-50, 1-crop 224×224**), trained on features from unsupervised pre-training. "aug+" in SimCLR includes blur and stronger color distortion. SimCLR ablations are from Fig. 9 in [2] (we thank the authors for providing the numerical results).

- Key takeaways:
- Non-linear projection head and strong data augmentation are crucial for contrastive learning.
- Decoupling mini-batch size with negative sample size allows MoCo-V2 to outperform SimCLR with smaller batch size (256 vs. 8192).

Instance vs. Sequence Contrastive Learning



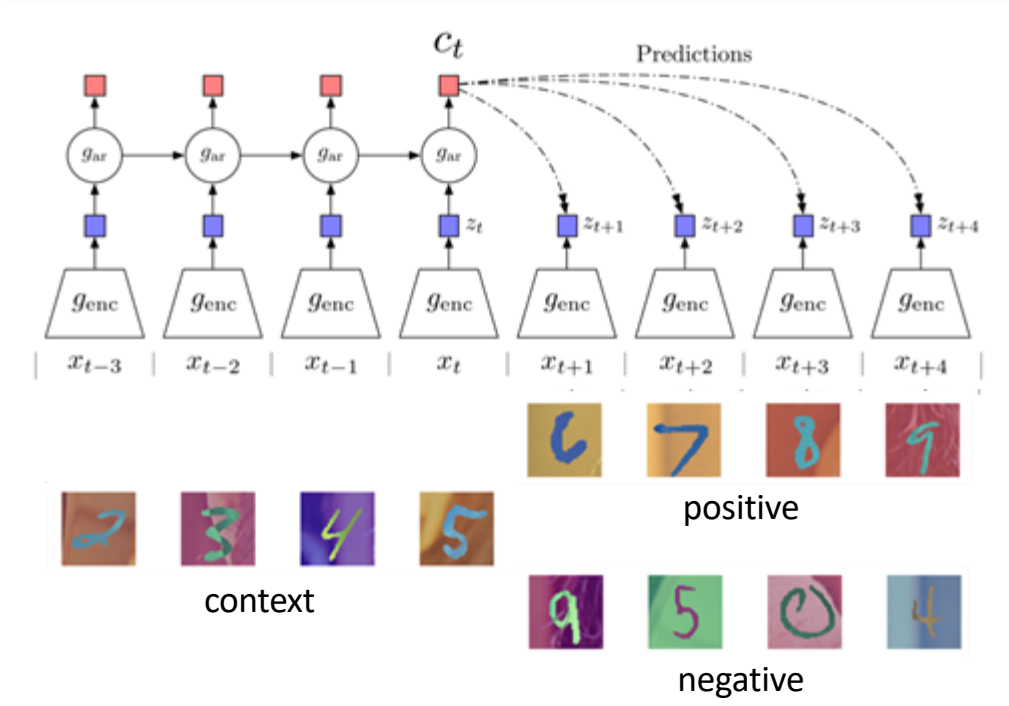


Source: van den Oord et al., 2018

Instance-level contrastive learning:
contrastive learning based on
positive & negative instances.
Examples: SimCLR, MoCo

Sequence-level contrastive learning:
contrastive learning based on
sequential / temporal orders.
Example: Contrastive Predictive Coding (CPC)

Contrastive Predictive Coding (CPC)



Representation learning with **contrastive predictive coding**
A Oord, Y Li, O Vinyals - arXiv preprint arXiv:1807.03748, 2018 - arxiv.org
… unsupervised **learning** approach to extract useful **representations** from high-… **Contrastive Predictive Coding**. The key insight of our model is to **learn** such **representations** by **predicting** …
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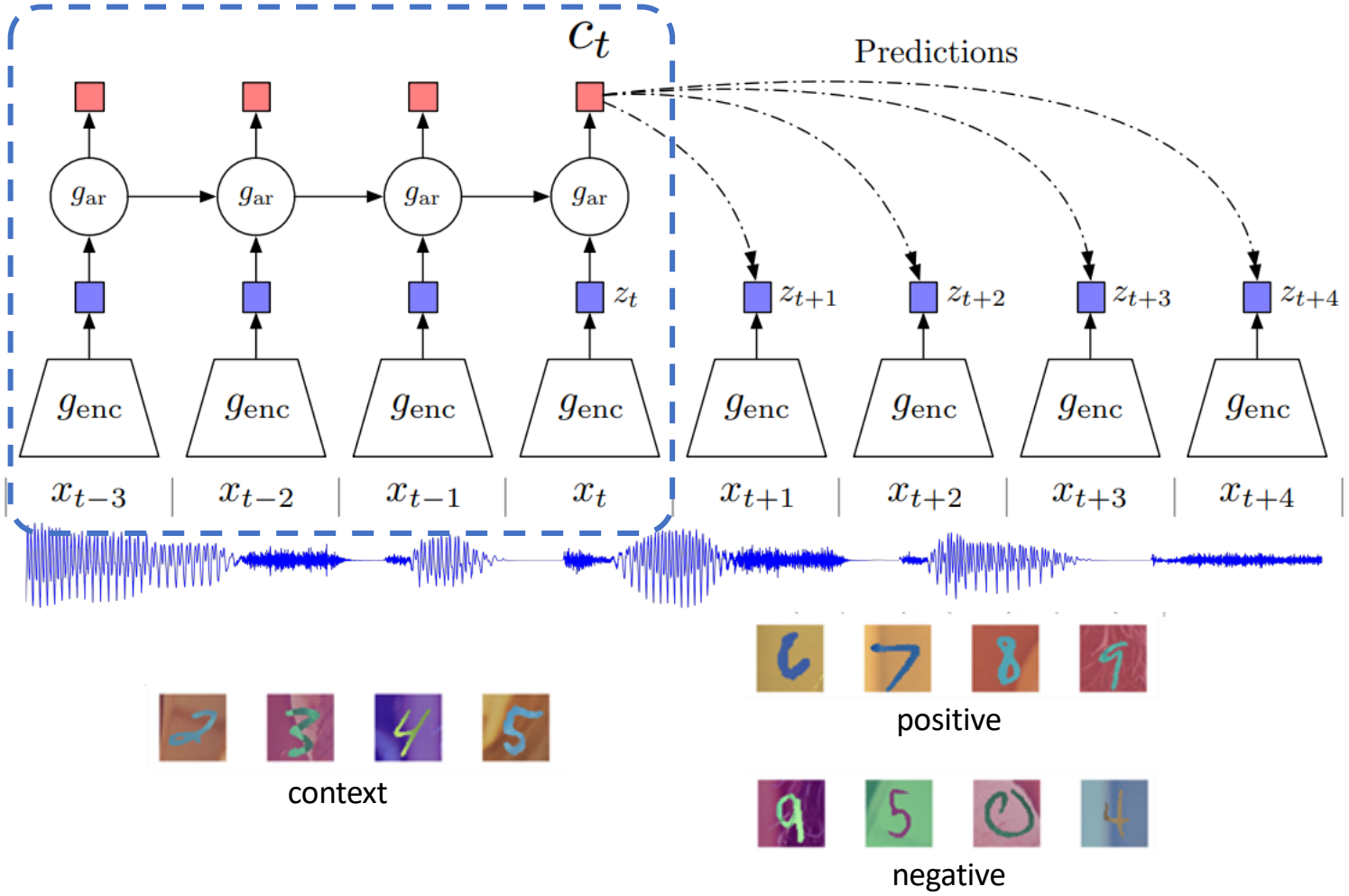
- Contrastive: contrast between "right" and "wrong" sequences using contrastive learning.
- Predictive: the model has to predict future patterns given the current context.
- Coding: the model learns useful feature vectors, or "code", for downstream tasks, similar to other self-supervised methods.

Contrastive Predictive Coding (CPC)



- Encode all samples in a sequence into vectors $z_t = g_{enc}(x_t)$
- Summarize context (e.g., half of a sequence) into a context code c_t using an auto-regressive model (g_{ar})
- Compute contrastive loss between the context c_t and future code z_{t+k} using the following time-dependent score function:

$$s_k(z_{t+k}, c_t) = z_{t+k}^T W_k c_t$$

where W_k is a trainable matrix

Contrastive Predictive Coding (CPC)

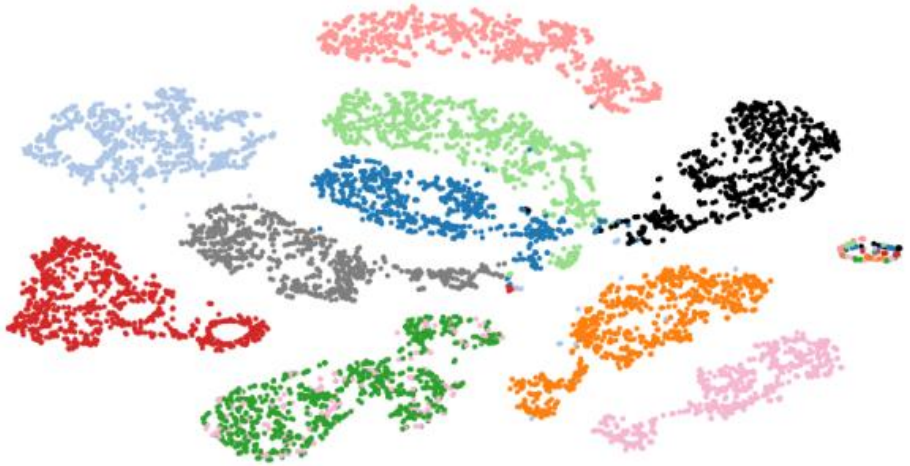


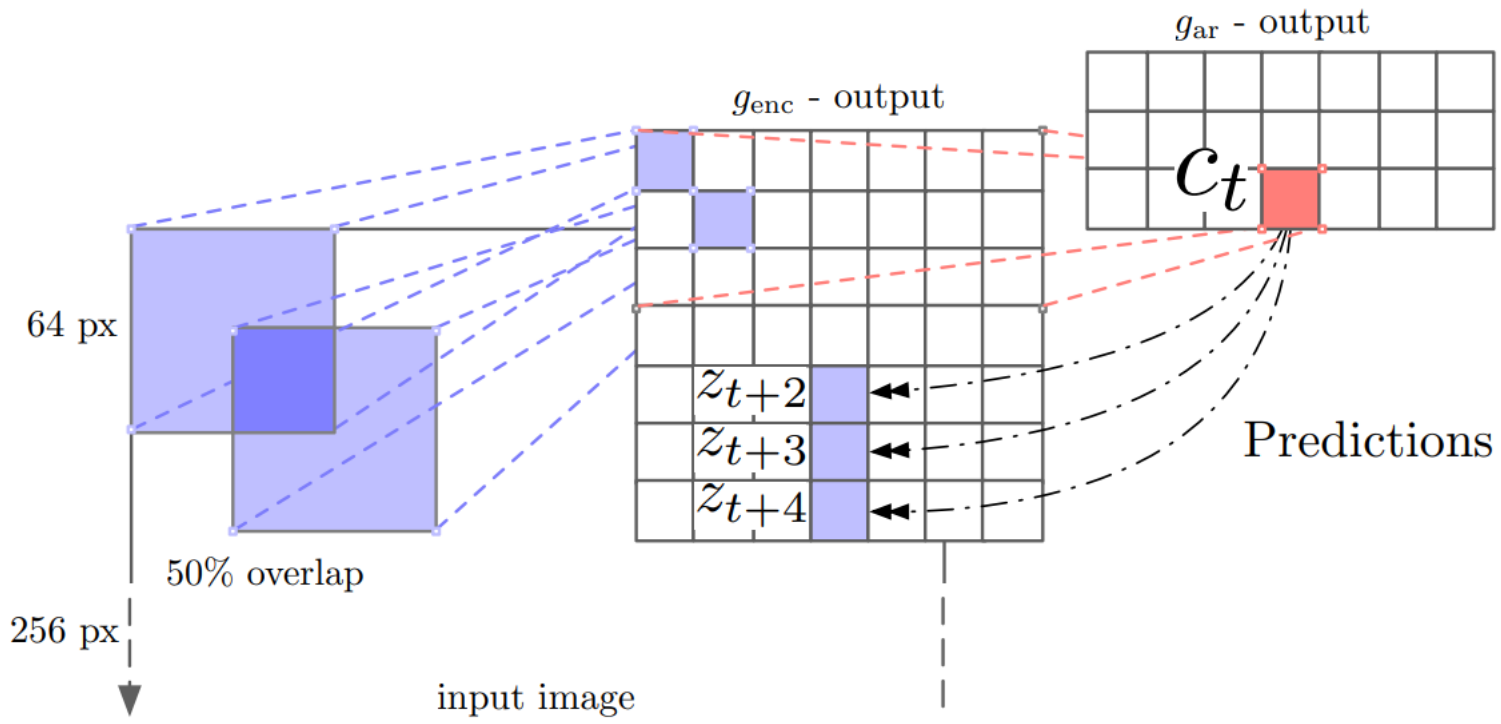
Figure 2: t-SNE visualization of audio (speech) representations for a subset of 10 speakers (out of 251). Every color represents a different speaker.

Method	ACC
Phone classification	
Random initialization	27.6
MFCC features	39.7
CPC	64.6
Supervised	74.6
Speaker classification	
Random initialization	1.87
MFCC features	17.6
CPC	97.4
Supervised	98.5

Table 1: LibriSpeech phone and speaker classification results. For phone classification there are 41 possible classes and for speaker classification 251. All models used the same architecture and the same audio input sizes.

CPC Example: modeling visual context

- Idea: split image into patches, model rows of patches from top to bottom as a sequence. I.e., use top rows as context to predict bottom rows.



DINO: Self-Distillation with No Labels

Emerging Properties in Self-Supervised Vision Transformers

Mathilde Caron[1,2] Hugo Touvron[1,3] Ishan Misra[1] Hervé Jegou[1]
Julien Mairal[2] Piotr Bojanowski[1] Armand Joulin[1]

[1] Facebook AI Research [2] Inria* [3] Sorbonne University

Figure 1: **Self-attention from a Vision Transformer with 8×8 patches trained with no supervision.** We look at the self-attention of the [CLS] token on the heads of the last layer. This token is not attached to any label nor supervision. These maps show that the model automatically learns class-specific features leading to unsupervised object segmentations.

Emerging properties in self-supervised vision transformers

M Caron, H Touvron, I Misra, H Jégou, J Mairal, P Bojanowski, A Joulin

Proceedings of the IEEE/CVF international conference on …, **2021** • openaccess.thecvf.com

Abstract

In this paper, we question if self-supervised learning provides new properties to Vision Transformer (ViT) that stand out compared to convolutional networks (convnets). Beyond the fact that adapting self-supervised methods to this architecture works particularly well, we make the following observations: first, self-supervised ViT features contain explicit information about the semantic segmentation of an image, which does not emerge as clearly with supervised ViTs, nor with convnets. Second, these features are also excellent

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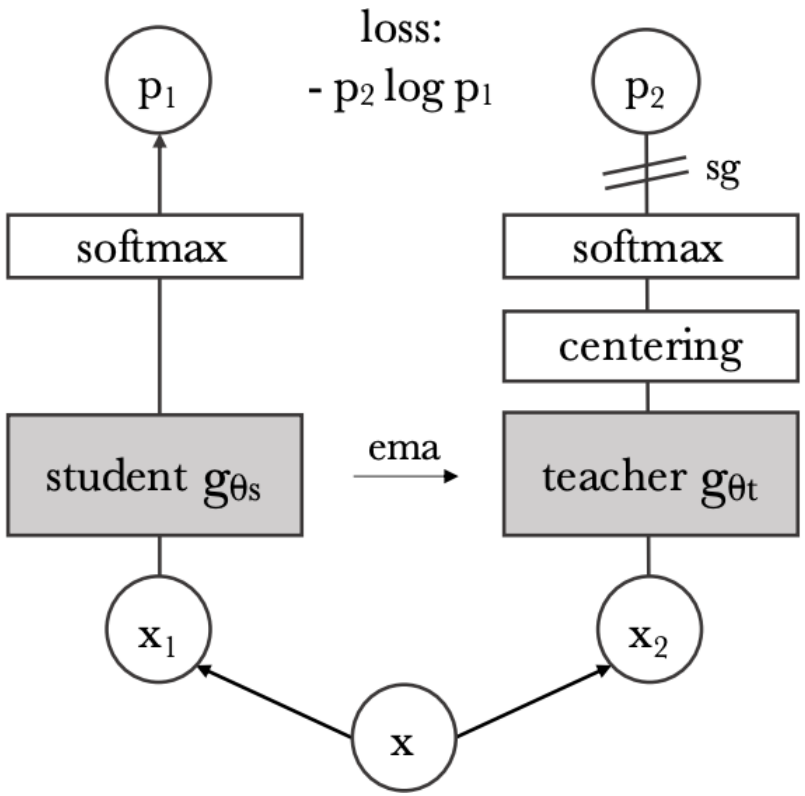
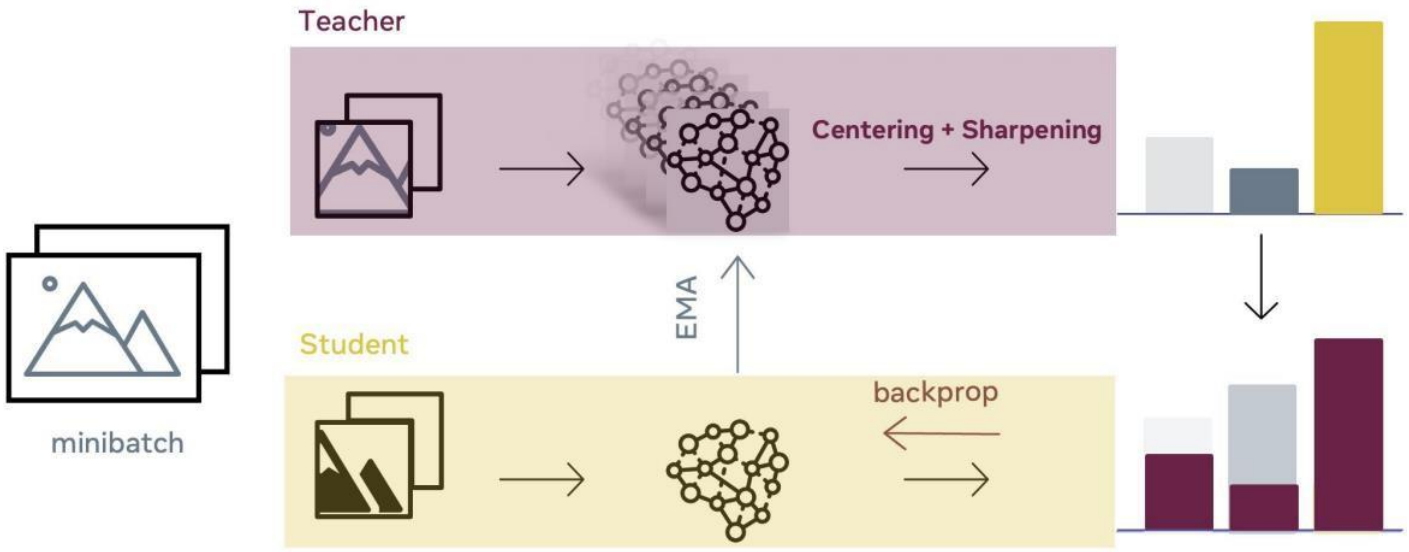
DINO

- Key idea:
  - No labels are used.
  - Uses a student–teacher (EMA) framework.
  - Learns strong visual representations by matching the output distributions of different augmented views of the same image.
  - Often built on Vision Transformers (ViT) or CNN backbones.
- Why it’s important:
  - Produces features that naturally cluster semantic objects.
  - Attention maps often highlight meaningful object regions without supervision.
  - Frequently used for: Feature extraction, Clustering, Transfer learning

DINO

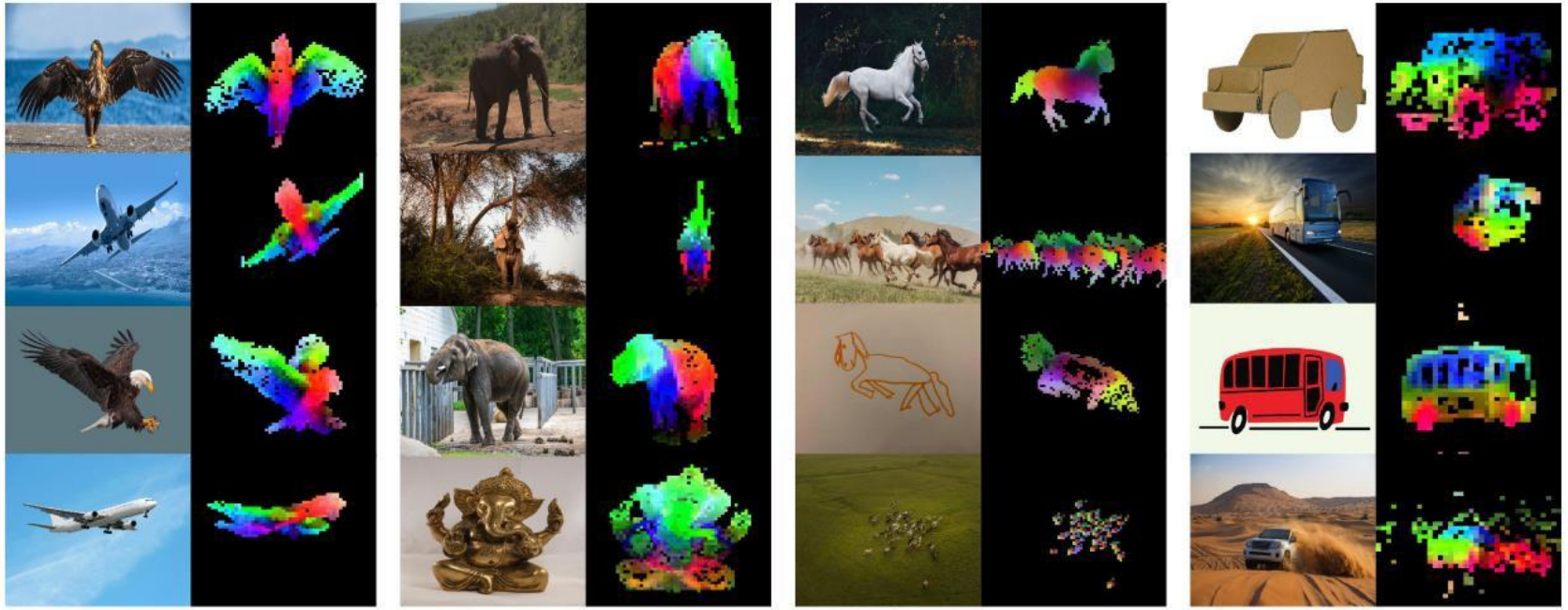




$$\theta_{\text{teacher}} \leftarrow \tau \cdot \theta_{\text{teacher}} + (1 - \tau) \cdot \theta_{\text{student}}$$

DINOv2

- Scales up training data from 1M ImageNet images to 142M images Very strong image features, commonly used in practice



PCA feature visualization

Oquab et al, “DINOv2: Learning Robust Visual Features without Supervision”, arXiv 2023; Darcet et al, “Vision Transformers Need Registers”, arXiv 2023

Summary

- Representation Learning
- Pretext tasks from image transformations
  - Rotation, inpainting, rearrangement, coloring
  - Reconstruction-based learning (MAE)
- Contrastive representation learning
  - Intuition and formulation
  - Instance contrastive learning: SimCLR and MOCO
  - Sequence contrastive learning: CPC
  - Self-Distillation Without Labels, DINO